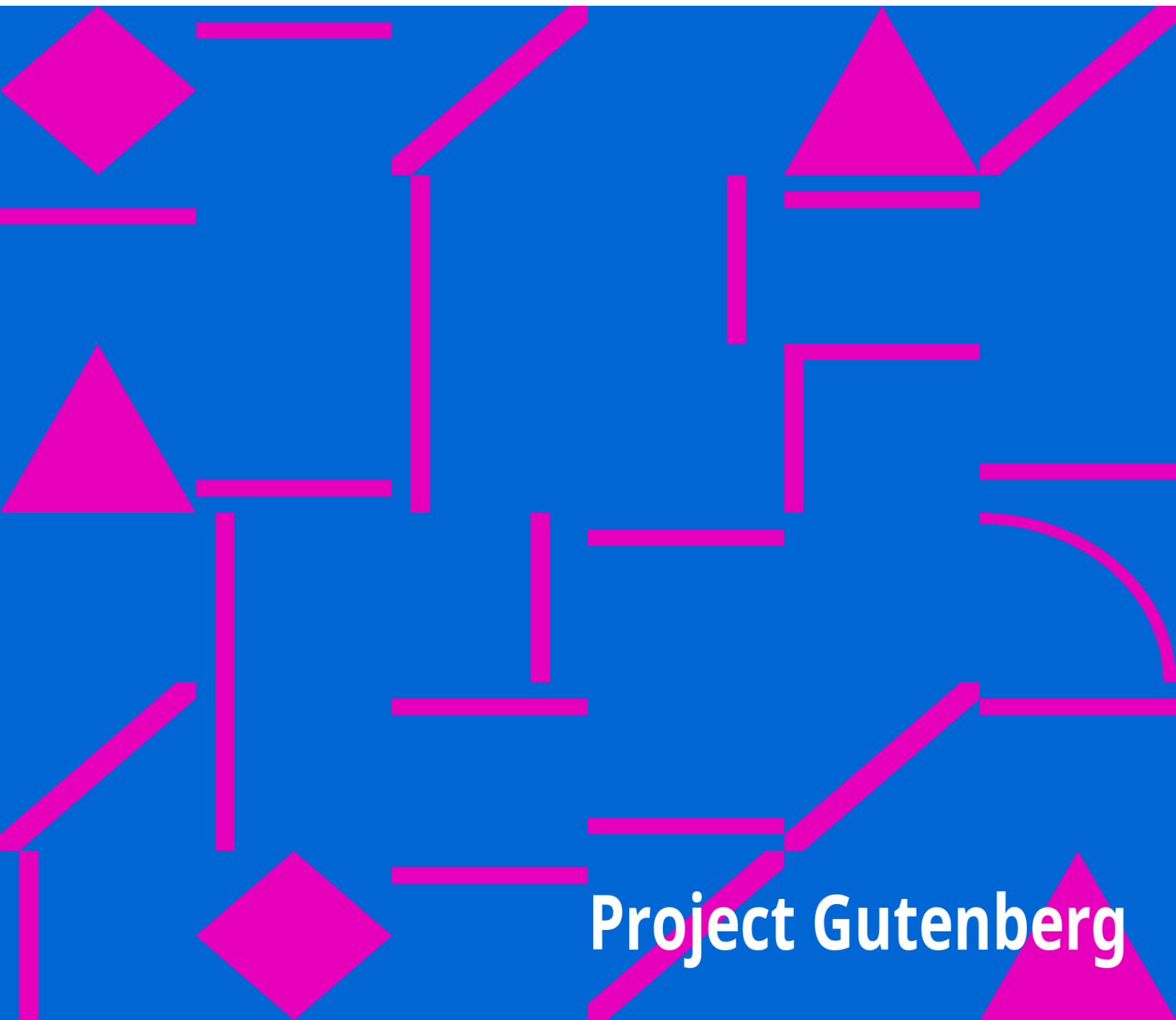


Zen and the Art of the Internet

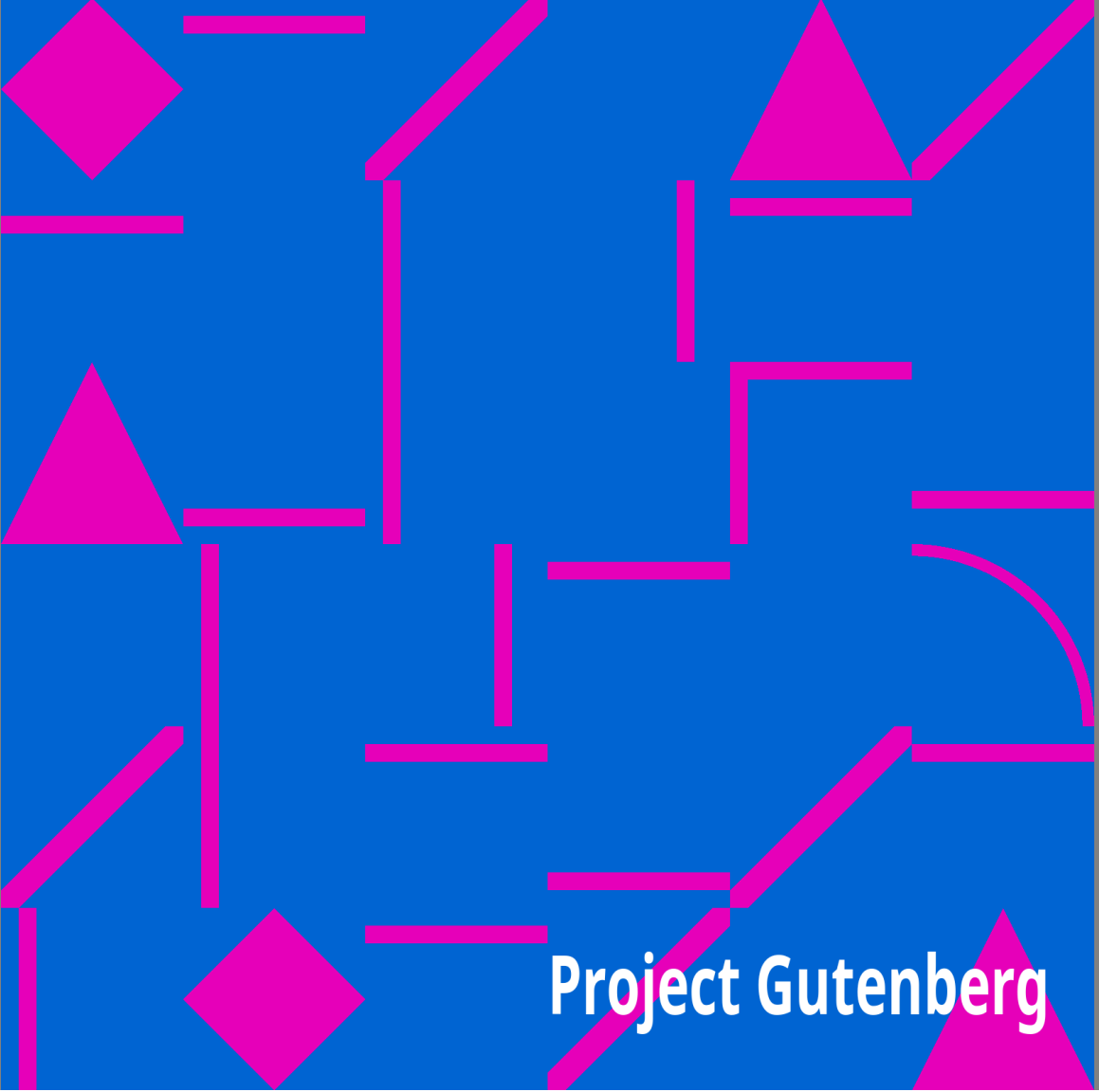
Brendan P. Kehoe

The background of the lower half of the image is a solid blue color. Overlaid on this background is a complex, abstract pattern of magenta geometric shapes. These shapes include various sizes of triangles, diamonds, squares, and lines, some of which are oriented diagonally. The pattern is dense and non-repeating, creating a modern, geometric aesthetic.

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Zen and the Art of the Internet

Brendan P. Kehoe

The lower half of the book cover features a vibrant blue background adorned with a complex, abstract pattern of magenta geometric shapes. These shapes include various sizes of triangles, diamonds, squares, and lines, some of which are interconnected to form larger, more intricate structures. The pattern is dense and covers the entire lower section of the cover.

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Part A

Zen and the Art of the Internet

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Zen and the Art of the Internet
A Beginner's Guide to the Internet
First Edition
January 1992

by Brendan P. Kehoe

This is revision 1.0 of February 2, 1992.
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The composition of this booklet was originally started because the Computer Science department at Widener University was in desperate need of documentation describing the capabilities of this "great new Internet link" we obtained.

It's since grown into an effort to acquaint the reader with much of what's currently available over the Internet. Aimed at the novice user, it attempts to remain operating system "neutral"—little information herein is specific to Unix, VMS, or any other environment. This booklet will, hopefully, be usable by nearly anyone.

A user's session is usually offset from the rest of the paragraph, as such:

prompt> command The results are usually displayed here.

The purpose of this booklet is two-fold: first, it's intended to serve as a reference piece, which someone can easily grab on the fly and look something up. Also, it forms a foundation from which people can explore the vast expanse of the Internet. Zen and the Art of the Internet doesn't spend a significant amount of time on any one point; rather, it provides enough for people to learn the specifics of what his or her local system offers.

One warning is perhaps in order—this territory we are entering can become a fantastic time-sink. Hours can slip by, people can come and go, and you'll be locked into Cyberspace. Remember to do your work!

With that, I welcome you, the new user, to The Net.

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Acknowledgements

Certain sections in this booklet are not my original work—rather, they are derived from documents that were available on the Internet and already aptly stated their areas of concentration. The chapter on Usenet is, in large part, made up of what's posted monthly to news.announce.newusers, with some editing and rewriting. Also, the main section on archie was derived from whatis.archie by Peter Deutsch of the McGill University Computing Centre. It's available via anonymous FTP from archie.mcgill.ca. Much of what's in the telnet section came from an impressive introductory document put together by SuraNet. Some definitions in the one are from an excellent glossary put together by Colorado State University.

This guide would not be the same without the aid of many people on The Net, and the providers of resources that are already out there. I'd like to thank the folks who gave this a read-through and returned some excellent comments, suggestions, and criticisms, and those who provided much-needed information on the fly. Glee Willis deserves particular mention for all of his work; this guide would have been considerably less polished without his help.

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Network Basics

We are truly in an information society. Now more than ever, moving vast amounts of information quickly across great distances is one of our most pressing needs. From small one-person entrepreneurial efforts, to the largest of corporations, more and more professional people are discovering that the only way to be successful in the '90s and beyond is to realize that technology is advancing at a break-neck pace—and they must somehow keep up. Likewise, researchers from all corners of the earth are finding that their work thrives in a networked environment. Immediate access to the work of colleagues and a "virtual" library of millions of volumes and thousands of papers affords them the ability to incorporate a body of knowledge heretofore unthinkable. Work groups can now conduct interactive conferences with each other, paying no heed to physical location—the possibilities are endless.

You have at your fingertips the ability to talk in "real-time" with someone in Japan, send a 2,000-word short story to a group of people who will critique it for the sheer pleasure of doing so, see if a Macintosh sitting in a lab in Canada is turned on, and find out if someone happens to be sitting in front of their computer (logged on) in Australia, all inside of thirty minutes. No airline (or tardis, for that matter) could ever match that travel itinerary.

The largest problem people face when first using a network is grasping all that's available. Even seasoned users find themselves surprised when they discover a new service or feature that they'd never known even existed. Once acquainted with the terminology and sufficiently comfortable with making occasional mistakes, the learning process will drastically speed up.

Domains

Getting where you want to go can often be one of the more difficult aspects of using networks. The variety of ways that places are named will probably leave a blank stare on your face at first. Don't fret; there is a method to this apparent madness.

If someone were to ask for a home address, they would probably expect a street, apartment, city, state, and zip code. That's all the information the post office needs to deliver mail in a reasonably speedy fashion. Likewise, computer addresses have a structure to them. The general form is:

a person's email address on a computer: user@somewhere.domain
a computer's name: somewhere.domain

The user portion is usually the person's account name on the system, though it doesn't have to be. somewhere.domain tells you the name of a system or location, and what kind of organization it is. The trailing domain is often one of the following:

com Usually a company or other commercial institution or organization, like Convex Computers (convex.com).

edu An educational institution, e.g. New York University, named nyu.edu.

gov A government site; for example, NASA is nasa.gov.

mil A military site, like the Air Force (af.mil).

net Gateways and other administrative hosts for a network (it does not mean all of the hosts in a network). {The Matrix, 111. One such gateway is near.net.}

org

This is a domain reserved for private organizations, who don't comfortably fit in the other classes of domains. One example is the Electronic Frontier Foundation named eff.org.

Each country also has its own top-level domain. For example, the us domain includes each of the fifty states. Other countries represented with domains include:

au Australia ca Canada fr France uk The United Kingdom. These also have sub-domains of things like ac.uk for academic sites and co.uk for commercial ones.

FQDN (Fully Qualified Domain Name)

The proper terminology for a site's domain name (somewhere.domain above) is its Fully Qualified Domain Name (FQDN). It is usually selected to give a clear indication of the site's organization or sponsoring agent. For example, the Massachusetts Institute of Technology's FQDN is mit.edu; similarly, Apple Computer's domain name is apple.com. While such obvious names are usually the norm, there are the occasional exceptions that are ambiguous enough to mislead—like vt.edu, which on first impulse one might surmise is an educational institution of some sort in Vermont; not so. It's actually the domain name for Virginia Tech. In most cases it's relatively easy to glean the meaning of a domain name—such confusion is far from the norm.

Internet Numbers

Every single machine on the Internet has a unique address, {At least one address, possibly two or even three—but we won't go into that.} called its Internet number or IP Address. It's actually a 32-bit number, but is most commonly represented as four numbers joined by periods (.), like 147.31.254.130. This is sometimes also called a dotted quad; there are literally thousands of different possible dotted quads. The ARPAnet (the mother to today's Internet) originally only had the capacity to have up to 256 systems on it because of the way each system was addressed. In the early eighties, it became clear that things would fast outgrow such a small limit; the 32-bit addressing method was born, freeing thousands of host numbers.

Each piece of an Internet address (like 192) is called an "octet," representing one of four sets of eight bits. The first two or three pieces (e.g. 192.55.239) represent the network that a system is on, called its subnet. For example, all of the computers for Wesleyan University are in the subnet 129.133. They can have numbers like 129.133.10.10, 129.133.230.19, up to 65 thousand possible combinations (possible computers).

IP addresses and domain names aren't assigned arbitrarily—that would lead to unbelievable confusion. An application must be filed with the Network Information Center (NIC), either electronically (to hostmaster@nic.ddn.mil) or via regular mail.

Resolving Names and Numbers

Ok, computers can be referred to by either their FQDN or their Internet address. How can one user be expected to remember them all?

They aren't. The Internet is designed so that one can use either method. Since humans find it much more natural to deal with words than numbers in most cases, the FQDN for each host is mapped to its Internet number. Each domain is served by a computer within that domain, which provides all of the necessary information to go from a domain name to an IP address, and vice-versa. For example, when someone refers to foosun.bar.com, the resolver knows that it should ask the system foovax.bar.com about systems in bar.com. It asks what Internet address foosun.bar.com has; if the name foosun.bar.com really exists, foovax will send back its number. All of this "magic" happens behind the scenes.

Rarely will a user have to remember the Internet number of a site (although often you'll catch yourself remembering an apparently obscure number, simply because you've accessed the system frequently). However, you will remember a substantial number of FQDNs. It will eventually reach a point when you are able to make a reasonably accurate guess at what domain name a certain college, university, or company might have, given just their name.

The Networks

Internet The Internet is a large "network of networks." There is no one network known as The Internet; rather, regional nets like SuraNet, PrepNet, NearNet, et al., are all inter-connected (nay, "inter-networked") together into one great living thing, communicating at amazing speeds with the TCP/IP protocol. All activity takes place in "real-time."

UUCP The UUCP network is a loose association of systems all communicating with the UUCP protocol. (UUCP stands for 'Unix-to-Unix Copy Program'.) It's based on two systems connecting to each other at specified intervals, called polling, and executing any work scheduled for

either of them. Historically most UUCP was done with Unix equipment, although the software's since been implemented on other platforms (e.g. VMS). For example, the system oregano polls the system basil once every two hours. If there's any mail waiting for oregano, basil will send it at that time; likewise, oregano will at that time send any jobs waiting for basil.

BITNET BITNET (the "Because It's Time Network") is comprised of systems connected by point-to-point links, all running the NJE protocol. It's continued to grow, but has found itself suffering at the hands of the falling costs of Internet connections. Also, a number of mail gateways are in place to reach users on other networks.

The Physical Connection

The actual connections between the various networks take a variety of forms. The most prevalent for Internet links are 56k leased lines (dedicated telephone lines carrying 56kilobit-per-second connections) and T1 links (special phone lines with 1Mbps connections). Also installed are T3 links, acting as backbones between major locations to carry a massive 45Mbps load of traffic.

These links are paid for by each institution to a local carrier (for example, Bell Atlantic owns PrepNet, the main provider in Pennsylvania). Also available are SLIP connections, which carry Internet traffic (packets) over high-speed modems.

UUCP links are made with modems (for the most part), that run from 1200 baud all the way up to as high as 38.4Kbps. As was mentioned in The Networks, the connections are of the store-and-forward variety. Also in use are Internet-based UUCP links (as if things weren't already confusing enough!). The systems do their UUCP traffic over TCP/IP connections, which give the UUCP-based network some blindingly fast "hops," resulting

in better connectivity for the network as a whole. UUCP connections first became popular in the 1970's, and have remained in wide-spread use ever since. Only with UUCP can Joe Smith correspond with someone across the country or around the world, for the price of a local telephone call.

BITNET links mostly take the form of 9600bps modems connected from site to site. Often places have three or more links going; the majority, however, look to "upstream" sites for their sole link to the network.

"The Glory and the Nothing of a Name"
Byron, {Churchill's Grave}

Electronic Mail

The desire to communicate is the essence of networking. People have always wanted to correspond with each other in the fastest way possible, short of normal conversation. Electronic mail (or email) is the most prevalent application of this in computer networking. It allows people to write back and forth without having to spend much time worrying about how the message actually gets delivered. As technology grows closer and closer to being a common part of daily life, the need to understand the many ways it can be utilized and how it works, at least to some level, is vital. part of daily life (as has been evidenced by the ISDN effort, the need to understand the many ways it can be utilized and how it works, at least to some level, is vital.

Email Addresses

Electronic mail is hinged around the concept of an address; the section on Networking Basics made some reference to it while introducing domains. Your email address provides all of the information required to get a

message to you from anywhere in the world. An address doesn't necessarily have to go to a human being. It could be an archive server, {See Archive Servers, for a description.} a list of people, or even someone's pocket pager. These cases are the exception to the norm—mail to most addresses is read by human beings.

%@!.: Symbolic Cacophony

Email addresses usually appear in one of two forms—using the Internet format which contains @, an "at"-sign, or using the UUCP format which contains !, an exclamation point, also called a "bang." The latter of the two, UUCP "bang" paths, is more restrictive, yet more clearly dictates how the mail will travel.

To reach Jim Morrison on the system south.america.org, one would address the mail as jm@south.america.org. But if Jim's account was on a UUCP site named brazil, then his address would be brazil!jm. If it's possible (and one exists), try to use the Internet form of an address; bang paths can fail if an intermediate site in the path happens to be down. There is a growing trend for UUCP sites to register Internet domain names, to help alleviate the problem of path failures.

Another symbol that enters the fray is %—it acts as an extra "routing" method. For example, if the UUCP site dream is connected to south.america.org, but doesn't have an Internet domain name of its own, a user debbie on dream can be reached by writing to the address not smallexample!

debbie%dream@south.america.org

The form is significant. This address says that the local system should first send the mail to south.america.org. There the address debbie%dream

will turn into debbie@dream, which will hopefully be a valid address. Then south.america.org will handle getting the mail to the host dream, where it will be delivered locally to debbie.

All of the intricacies of email addressing methods are fully covered in the book "!"%@@:: A Directory of Electronic Mail Addressing and Networks" published by O'Reilly and Associates, as part of their Nutshell Handbook series. It is a must for any active email user. Write to nuts@ora.com for ordering information.

Sending and Receiving Mail

We'll make one quick diversion from being OS-neuter here, to show you what it will look like to send and receive a mail message on a Unix system. Check with your system administrator for specific instructions related to mail at your site.

A person sending the author mail would probably do something like this:

```
% mail brendan@cs.widener.edu Subject: print job's stuck
```

I typed `print babe.gif' and it didn't work! Why??

The next time the author checked his mail, he would see it listed in his mailbox as:

```
% mail "/usr/spool/mail/brendan": 1 messages 1 new 1 unread U 1  
joeuser@foo.widene Tue May 5 20:36 29/956 print job's stuck ?
```

which gives information on the sender of the email, when it was sent, and the subject of the message. He would probably use the reply command of Unix mail to send this response:

? r

To: joeuser@@foo.widener.edu

Subject: Re: print job's stuck

You shouldn't print binary files like GIFs to a printer!

Brendan

Try sending yourself mail a few times, to get used to your system's mailer. It'll save a lot of wasted aspirin for both you and your system administrator.

Anatomy of a Mail Header

An electronic mail message has a specific structure to it that's common across every type of computer system. {The standard is written down in RFC-822. See also RFCs for more info on how to get copies of the various RFCs.} A sample would be:

```
>From bush@hq.mil Sat May 25 17:06:01 1991
Received: from hq.mil by house.gov with SMTP id AA21901
(4.1/SMI for dan@house.gov); Sat, 25 May 91 17:05:56 -0400
Date: Sat, 25 May 91 17:05:56 -0400
From: The President <bush@hq.mil>
Message-Id: <9105252105.AA06631@hq.mil>
To: dan@senate.gov
Subject: Meeting
```

Hi Dan .. we have a meeting at 9:30 a.m. with the Joint Chiefs. Please don't oversleep this time.

The first line, with From and the two lines for Received: are usually not very interesting. They give the "real" address that the mail is coming from

(as opposed to the address you should reply to, which may look much different), and what places the mail went through to get to you. Over the Internet, there is always at least one Received: header and usually no more than four or five. When a message is sent using UUCP, one Received: header is added for each system that the mail passes through. This can often result in more than a dozen Received: headers. While they help with dissecting problems in mail delivery, odds are the average user will never want to see them. Most mail programs will filter out this kind of "cruft" in a header.

The Date: header contains the date and time the message was sent. Likewise, the "good" address (as opposed to "real" address) is laid out in the From: header. Sometimes it won't include the full name of the person (in this case The President), and may look different, but it should always contain an email address of some form.

The Message-ID: of a message is intended mainly for tracing mail routing, and is rarely of interest to normal users. Every Message-ID: is guaranteed to be unique.

To: lists the email address (or addresses) of the recipients of the message. There may be a Cc: header, listing additional addresses. Finally, a brief subject for the message goes in the Subject: header.

The exact order of a message's headers may vary from system to system, but it will always include these fundamental headers that are vital to proper delivery.

Bounced Mail

When an email address is incorrect in some way (the system's name is wrong, the domain doesn't exist, whatever), the mail system will bounce the

message back to the sender, much the same way that the Postal Service does when you send a letter to a bad street address. The message will include the reason for the bounce; a common error is addressing mail to an account name that doesn't exist. For example, writing to Lisa Simpson at Widener University's Computer Science department will fail, because she doesn't have an account. {Though if she asked, we'd certainly give her one.}

From: Mail Delivery Subsystem <MAILER-DAEMON>
Date: Sat, 25 May 91 16:45:14 -0400
To: mg@gracie.com
Cc: Postmaster@cs.widener.edu
Subject: Returned mail: User unknown

—— Transcript of session follows —— While talking to
cs.widener.edu: >>> RCPT To:<lsimpson@cs.widener.edu> <<< 550
<lsimpson@cs.widener.edu>... User unknown 550 lsimpson... User
unknown

As you can see, a carbon copy of the message (the Cc: header entry) was sent to the postmaster of Widener's CS department. The Postmaster is responsible for maintaining a reliable mail system on his system. Usually postmasters at sites will attempt to aid you in getting your mail where it's supposed to go. If a typing error was made, then try re-sending the message. If you're sure that the address is correct, contact the postmaster of the site directly and ask him how to properly address it.

The message also includes the text of the mail, so you don't have to retype everything you wrote.

—— Unsent message follows ——
Received: by cs.widener.edu id AA06528; Sat, 25 May 91 16:45:14 -0400
Date: Sat, 25 May 91 16:45:14 -0400

From: Matt Groening <mg@gracie.com>
Message-Id: <9105252045.AA06528@gracie.com>
To: lsimpson@cs.widener.edu
Subject: Scripting your future episodes
Reply-To: writing-group@gracie.com

.... verbiage ...

The full text of the message is returned intact, including any headers that were added. This can be cut out with an editor and fed right back into the mail system with a proper address, making redelivery a relatively painless process.

Mailing Lists

People that share common interests are inclined to discuss their hobby or interest at every available opportunity. One modern way to aid in this exchange of information is by using a mailing list—usually an email address that redistributes all mail sent to it back out to a list of addresses. For example, the Sun Managers mailing list (of interest to people that administer computers manufactured by Sun) has the address sun-managers@eecs.nwu.edu. Any mail sent to that address will "explode" out to each person named in a file maintained on a computer at Northwestern University.

Administrative tasks (sometimes referred to as administrivia) are often handled through other addresses, typically with the suffix -request. To continue the above, a request to be added to or deleted from the Sun Managers list should be sent to sun-managers-request@eecs.nwu.edu.

When in doubt, try to write to the -request version of a mailing list address first; the other people on the list aren't interested in your desire to

be added or deleted, and can certainly do nothing to expedite your request. Often if the administrator of a list is busy (remember, this is all peripheral to real jobs and real work), many users find it necessary to ask again and again, often with harsher and harsher language, to be removed from a list. This does nothing more than waste traffic and bother everyone else receiving the messages. If, after a reasonable amount of time, you still haven't succeeded to be removed from a mailing list, write to the postmaster at that site and see if they can help.

Exercise caution when replying to a message sent by a mailing list. If you wish to respond to the author only, make sure that the only address you're replying to is that person, and not the entire list. Often messages of the sort "Yes, I agree with you completely!" will appear on a list, boring the daylights out of the other readers. Likewise, if you explicitly do want to send the message to the whole list, you'll save yourself some time by checking to make sure it's indeed headed to the whole list and not a single person.

A list of the currently available mailing lists is available in at least two places; the first is in a file on <ftp.nisc.sri.com> called `interest-groups` under the `netinfo/` directory. It's updated fairly regularly, but is large (presently around 700K), so only get it every once in a while. The other list is maintained by Gene Spafford (spaf@cs.purdue.edu), and is posted in parts to the newsgroup `news.lists` semi-regularly. (Usenet News, for info on how to read that and other newsgroups.)

Listservs

On BITNET there's an automated system for maintaining discussion lists called the `listserv`. Rather than have an already harried and overworked human take care of additions and removals from a list, a program performs these and other tasks by responding to a set of user-driven commands.

Areas of interest are wide and varied—ETHICS-L deals with ethics in computing, while ADND-L has to do with a role-playing game. A full list of the available BITNET lists can be obtained by writing to `LISTSERV@BITNIC.BITNET` with a body containing the command

```
list global
```

However, be sparing in your use of this—see if it's already on your system somewhere. The reply is quite large.

The most fundamental command is `subscribe`. It will tell the listserv to add the sender to a specific list. The usage is

```
subscribe foo-l Your Real Name
```

It will respond with a message either saying that you've been added to the list, or that the request has been passed on to the system on which the list is actually maintained.

The mate to `subscribe` is, naturally, `unsubscribe`. It will remove a given address from a BITNET list. It, along with all other listserv commands, can be abbreviated—`subscribe` as `sub`, `unsubscribe` as `unsub`, etc. For a full list of the available listserv commands, write to `LISTSERV@BITNIC.BITNET`, giving it the command `help`.

As an aside, there have been implementations of the listserv system for non-BITNET hosts (more specifically, Unix systems). One of the most complete is available on `cs.bu.edu` in the directory `pub/listserv`.

"I made this letter longer than usual because
I lack the time to make it shorter."
Pascal, Provincial Letters XVI

Anonymous FTP

FTP (File Transfer Protocol) is the primary method of transferring files over the Internet. On many systems, it's also the name of the program that implements the protocol. Given proper permission, it's possible to copy a file from a computer in South Africa to one in Los Angeles at very fast speeds (on the order of 5—10K per second). This normally requires either a user id on both systems or a special configuration set up by the system administrator(s).

There is a good way around this restriction—the anonymous FTP service. It essentially will let anyone in the world have access to a certain area of disk space in a non-threatening way. With this, people can make files publicly available with little hassle. Some systems have dedicated entire disks or even entire computers to maintaining extensive archives of source code and information. They include gatekeeper.dec.com (Digital), wuarchive.wustl.edu (Washington University in Saint Louis), and archive.cis.ohio-state.edu (The Ohio State University).

The process involves the "foreign" user (someone not on the system itself) creating an FTP connection and logging into the system as the user anonymous, with an arbitrary password:

Name (foo.site.com:you): anonymous
Password: jm@south.america.org

Custom and netiquette dictate that people respond to the Password: query with an email address so that the sites can track the level of FTP usage, if they desire. (Addresses for information on email addresses).

The speed of the transfer depends on the speed of the underlying link. A site that has a 9600bps SLIP connection will not get the same throughput as a system with a 56k leased line (The Physical Connection, for more on what kinds of connections can exist in a network). Also, the traffic of all other users on that link will affect performance. If there are thirty people all FTPing from one site simultaneously, the load on the system (in addition to the network connection) will degrade the overall throughput of the transfer.

FTP Etiquette

Lest we forget, the Internet is there for people to do work. People using the network and the systems on it are doing so for a purpose, whether it be research, development, whatever. Any heavy activity takes away from the overall performance of the network as a whole.

The effects of an FTP connection on a site and its link can vary; the general rule of thumb is that any extra traffic created detracts from the ability of that site's users to perform their tasks. To help be considerate of this, it's highly recommended that FTP sessions be held only after normal business hours for that site, preferably late at night. The possible effects of a large transfer will be less destructive at 2 a.m. than 2 p.m. Also, remember that if it's past dinner time in Maine, it's still early afternoon in California—think in terms of the current time at the site that's being visited, not of local time.

Basic Commands

While there have been many extensions to the various FTP clients out there, there is a de facto "standard" set that everyone expects to work. For more specific information, read the manual for your specific FTP program. This section will only skim the bare minimum of commands needed to operate an FTP session.

Creating the Connection

The actual command to use FTP will vary among operating systems; for the sake of clarity, we'll use FTP here, since it's the most general form.

There are two ways to connect to a system—using its hostname or its Internet number. Using the hostname is usually preferred. However, some sites aren't able to resolve hostnames properly, and have no alternative. We'll assume you're able to use hostnames for simplicity's sake. The form is

```
ftp somewhere.domain
```

Domains for help with reading and using domain names (in the example below, somewhere.domain is ftp.uu.net).

You must first know the name of the system you want to connect to. We'll use ftp.uu.net as an example. On your system, type:

```
ftp ftp.uu.net
```

(the actual syntax will vary depending on the type of system the connection's being made from). It will pause momentarily then respond with the message

```
Connected to ftp.uu.net.
```

and an initial prompt will appear:

```
220 uunet FTP server (Version 5.100 Mon Feb 11 17:13:28 EST 1991)
ready. Name (ftp.uu.net:jm):
```

to which you should respond with anonymous:

220 uunet FTP server (Version 5.100 Mon Feb 11 17:13:28 EST 1991)
ready. Name (ftp.uu.net:jm): anonymous

The system will then prompt you for a password; as noted previously, a good response is your email address:

331 Guest login ok, send ident as password. Password:
jm@south.america.org 230 Guest login ok, access restrictions apply. ftp>

The password itself will not echo. This is to protect a user's security when he or she is using a real account to FTP files between machines. Once you reach the ftp> prompt, you know you're logged in and ready to go.

Notice the ftp.uu.net:joe in the Name: prompt? That's another clue that anonymous FTP is special: FTP expects a normal user accounts to be used for transfers.

dir At the ftp> prompt, you can type a number of commands to perform various functions. One example is dir—it will list the files in the current directory. Continuing the example from above:

ftp> dir

200 PORT command successful. 150 Opening ASCII mode data
connection for /bin/ls. total 3116 drwxr-xr-x 2 7 21 512 Nov 21 1988
.forward -rw-rw-r— 1 7 11 0 Jun 23 1988 .hushlogin drwxrwxr-x 2 0 21
512 Jun 4 1990 Census drwxrwxr-x 2 0 120 512 Jan 8 09:36 ClariNet ... etc
etc ... -rw-rw-r— 1 7 14 42390 May 20 02:24 newthisweek.Z ... etc etc ...
-rw-rw-r— 1 7 14 2018887 May 21 01:01 uumap.tar.Z drwxrwxr-x 2 7 6
1024 May 11 10:58 uunet-info

226 Transfer complete. 5414 bytes received in 1.1 seconds (4.9 Kbytes/s)
ftp>

The file newthisweek.Z was specifically included because we'll be using it later. Just for general information, it happens to be a listing of all of the files added to UUNET's archives during the past week.

The directory shown is on a machine running the Unix operating system—the `dir` command will produce different results on other operating systems (e.g. TOPS, VMS, et al.). Learning to recognize different formats will take some time. After a few weeks of traversing the Internet, it proves easier to see, for example, how large a file is on an operating system you're otherwise not acquainted with.

With many FTP implementations, it's also possible to take the output of `dir` and put it into a file on the local system with

```
ftp> dir n* outfilename
```

the contents of which can then be read outside of the live FTP connection; this is particularly useful for systems with very long directories (like ftp.uu.net). The above example would put the names of every file that begins with an `n` into the local file `outfilename`.

```
cd
```

At the beginning of an FTP session, the user is in a "top-level" directory. Most things are in directories below it (e.g. `/pub`). To change the current directory, one uses the `cd` command. To change to the directory `pub`, for example, one would type

```
ftp> cd pub
```

which would elicit the response

```
250 CWD command successful.
```

Meaning the "Change Working Directory" command (cd) worked properly. Moving "up" a directory is more system-specific—in Unix use the command `cd ..`, and in VMS, `cd [-]`.

get and put

The actual transfer is performed with the get and put commands. To get a file from the remote computer to the local system, the command takes the form:

```
ftp> get filename
```

where filename is the file on the remote system. Again using ftp.uu.net as an example, the file newthisweek.Z can be retrieved with

```
ftp> get newthisweek.Z 200 PORT command successful. 150 Opening
ASCII mode data connection for newthisweek.Z (42390 bytes). 226
Transfer complete. local: newthisweek.Z remote: newthisweek.Z 42553
bytes received in 6.9 seconds (6 Kbytes/s) ftp>
```

The section below on using binary mode instead of ASCII will describe why this particular choice will result in a corrupt and subsequently unusable file.

If, for some reason, you want to save a file under a different name (e.g. your system can only have 14-character filenames, or can only have one dot in the name), you can specify what the local filename should be by providing get with an additional argument

```
ftp> get newthisweek.Z uunet-new
```

which will place the contents of the file newthisweek.Z in uunet-new on the local system.

The transfer works the other way, too. The put command will transfer a file from the local system to the remote system. If the permissions are set up for an FTP session to write to a remote directory, a file can be sent with

```
ftp> put filename
```

As with get, put will take a third argument, letting you specify a different name for the file on the remote system.

ASCII vs Binary

In the example above, the file newthisweek.Z was transferred, but supposedly not correctly. The reason is this: in a normal ASCII transfer (the default), certain characters are translated between systems, to help make text files more readable. However, when binary files (those containing non-ASCII characters) are transferred, this translation should not take place. One example is a binary program—a few changed characters can render it completely useless.

To avoid this problem, it's possible to be in one of two modes—ASCII or binary. In binary mode, the file isn't translated in any way. What's on the remote system is precisely what's received. The commands to go between the two modes are:

```
ftp> ascii 200 Type set to A. (Note the A, which signifies ASCII mode.)
```

```
ftp> binary 200 Type set to I. (Set to Image format, for pure binary transfers.)
```

Note that each command need only be done once to take effect; if the user types binary, all transfers in that session are done in binary mode (that is, unless ascii is typed later).

The transfer of newthisweek.Z will work if done as:

```
ftp> binary 200 Type set to I. ftp> get newthisweek.Z 200 PORT  
command successful. 150 Opening BINARY mode data connection for  
newthisweek.Z (42390 bytes). 226 Transfer complete. local: newthisweek.Z  
remote: newthisweek.Z 42390 bytes received in 7.2 seconds (5.8 Kbytes/s)
```

Note: The file size (42390) is different from that done in ASCII mode (42553) bytes; and the number 42390 matches the one in the listing of UUNET's top directory. We can be relatively sure that we've received the file without any problems.

mget and mput

The commands mget and mput allow for multiple file transfers using wildcards to get several files, or a whole set of files at once, rather than having to do it manually one by one. For example, to get all files that begin with the letter f, one would type

```
ftp> mget f*
```

Similarly, to put all of the local files that end with .c:

```
ftp> mput *.c
```

Rather than reiterate what's been written a hundred times before, consult a local manual for more information on wildcard matching (every DOS manual, for example, has a section on it).

Normally, FTP assumes a user wants to be prompted for every file in a mget or mput operation. You'll often need to get a whole set of files and not

have each of them confirmed—you know they're all right. In that case, use the prompt command to turn the queries off.

```
ftp> prompt Interactive mode off.
```

Likewise, to turn it back on, the prompt command should simply be issued again.

Joe Granrose's List

Monthly, Joe Granrose (odin@pilot.njin.net) posts to Usenet (Usenet News) an extensive list of sites offering anonymous FTP service. It's available in a number of ways:

The Usenet groups comp.misc and comp.sources.wanted

Anonymous FTP from pilot.njin.net [128.6.7.38], in /pub/ftp-list.

Write to odin@pilot.njin.net with a Subject: line of listserv-request and a message body of send help. Please don't bother Joe with your requests—the server will provide you with the list.

The archie Server archie is always in lowercase

A group of people at McGill University in Canada got together and created a query system called archie. It was originally formed to be a quick and easy way to scan the offerings of the many anonymous FTP sites that are maintained around the world. As time progressed, archie grew to include other valuable services as well.

The archie service is accessible through an interactive telnet session, email queries, and command-line and X-window clients. The email responses can be used along with FTPmail servers for those not on the Internet. (FTP-by-Mail Servers, for info on using FTPmail servers.)

Using archie Today

Currently, archie tracks the contents of over 800 anonymous FTP archive sites containing over a million files stored across the Internet. Collectively, these files represent well over 50 gigabytes of information, with new entries being added daily.

The archie server automatically updates the listing information from each site about once a month. This avoids constantly updating the databases, which could waste network resources, yet ensures that the information on each site's holdings is reasonably up to date.

To access archie interactively, telnet to one of the existing servers. {See Telnet, for notes on using the telnet program.} They include

archie.ans.net (New York, USA) archie.rutgers.edu (New Jersey, USA)
archie.sura.net (Maryland, USA) archie.unl.edu (Nebraska, USA)
archie.mcgill.ca (the first Archie server, in Canada) archie.funet.fi (Finland)
archie.au (Australia) archie.doc.ic.ac.uk (Great Britain)

At the login: prompt of one of the servers, enter archie to log in. A greeting will be displayed, detailing information about ongoing work in the archie project; the user will be left at a archie> prompt, at which he may enter commands. Using help will yield instructions on using the prog command to make queries, set to control various aspects of the server's operation, et al. Type quit at the prompt to leave archie. Typing the query prog vine.tar.Z will yield a list of the systems that offer the source to the X-windows program vine; a piece of the information returned looks like:

Host ftp.uu.net (137.39.1.9)

Last updated 10:30 7 Jan 1992

Location: /packages/X/contrib
FILE rw-r—r— 15548 Oct 8 20:29 vine.tar.Z

Host nic.funet.fi (128.214.6.100)
Last updated 05:07 4 Jan 1992

Location: /pub/X11/contrib
FILE rw-rw-r— 15548 Nov 8 03:25 vine.tar.Z

archie Clients

There are two main-stream archie clients, one called (naturally enough) archie, the other xarchie (for X-Windows). They query the archie databases and yield a list of systems that have the requested file(s) available for anonymous FTP, without requiring an interactive session to the server. For example, to find the same information you tried with the server command prog, you could type

% archie vine.tar.Z
Host athene.uni-paderborn.de
Location: /local/X11/more_contrib
FILE -rw-r—r— 18854 Nov 15 1990 vine.tar.Z

Host emx.utexas.edu
Location: /pub/mnt/source/games
FILE -rw-r—r— 12019 May 7 1988 vine.tar.Z

Host export.lcs.mit.edu
Location: /contrib
FILE -rw-r—r— 15548 Oct 9 00:29 vine.tar.Z

Note that your system administrator may not have installed the archie clients yet; the source is available on each of the archie servers, in the

directory archie/clients.

Using the X-windows client is much more intuitive—if it's installed, just read its man page and give it a whirl. It's essential for the networked desktop.

Mailing archie

Users limited to email connectivity to the Internet should send a message to the address `archie@archie.mcgill.ca` with the single word `help` in the body of the message. An email message will be returned explaining how to use the email archie server, along with the details of using FTPmail. Most of the commands offered by the telnet interface can be used with the mail server.

The whatis database

In addition to offering access to anonymous FTP listings, archie also permits access to the whatis description database. It includes the names and brief synopses for over 3,500 public domain software packages, datasets and informational documents located on the Internet.

Additional whatis databases are scheduled to be added in the future. Planned offerings include listings for the names and locations of online library catalog programs, the names of publicly accessible electronic mailing lists, compilations of Frequently Asked Questions lists, and archive sites for the most popular Usenet newsgroups. Suggestions for additional descriptions or locations databases are welcomed and should be sent to the archie developers at `archie-l@cs.mcgill.ca`.

"Was f@"ur pl@"undern!"
("What a place to plunder!")

Gebhard Leberecht Bl@"ucher

Usenet News

Original from: chip@count.tct.com (Chip Salzenberg)

[Most recent change: 19 May 1991 by spaf@cs.purdue.edu (Gene Spafford)]

The first thing to understand about Usenet is that it is widely misunderstood. Every day on Usenet the "blind men and the elephant" phenomenon appears, in spades. In the opinion of the author, more flame wars (rabid arguments) arise because of a lack of understanding of the nature of Usenet than from any other source. And consider that such flame wars arise, of necessity, among people who are on Usenet. Imagine, then, how poorly understood Usenet must be by those outside!

No essay on the nature of Usenet can ignore the erroneous impressions held by many Usenet users. Therefore, this section will treat falsehoods first. Keep reading for truth. (Beauty, alas, is not relevant to Usenet.)

What Usenet Is

Usenet is the set of machines that exchange articles tagged with one or more universally-recognized labels, called newsgroups (or "groups" for short). (Note that the term newsgroup is correct, while area, base, board, bboard, conference, round table, SIG, etc. are incorrect. If you want to be understood, be accurate.)

The Diversity of Usenet

If the above definition of Usenet sounds vague, that's because it is. It is almost impossible to generalize over all Usenet sites in any non-trivial way.

Usenet encompasses government agencies, large universities, high schools, businesses of all sizes, home computers of all descriptions, etc.

Every administrator controls his own site. No one has any real control over any site but his own. The administrator gets his power from the owner of the system he administers. As long as the owner is happy with the job the administrator is doing, he can do whatever he pleases, up to and including cutting off Usenet entirely. C'est la vie.

What Usenet Is Not

Usenet is not an organization. Usenet has no central authority. In fact, it has no central anything. There is a vague notion of "upstream" and "downstream" related to the direction of high-volume news flow. It follows that, to the extent that "upstream" sites decide what traffic they will carry for their "downstream" neighbors, that "upstream" sites have some influence on their neighbors. But such influence is usually easy to circumvent, and heavy-handed manipulation typically results in a backlash of resentment.

Usenet is not a democracy. A democracy can be loosely defined as "government of the people, by the people, for the people." However, as explained above, Usenet is not an organization, and only an organization can be run as a democracy. Even a democracy must be organized, for if it lacks a means of enforcing the peoples' wishes, then it may as well not exist.

Some people wish that Usenet were a democracy. Many people pretend that it is. Both groups are sadly deluded.

Usenet is not fair. After all, who shall decide what's fair? For that matter, if someone is behaving unfairly, who's going to stop him? Neither you nor I,

that's certain.

Usenet is not a right. Some people misunderstand their local right of "freedom of speech" to mean that they have a legal right to use others' computers to say what they wish in whatever way they wish, and the owners of said computers have no right to stop them.

Those people are wrong. Freedom of speech also means freedom not to speak; if I choose not to use my computer to aid your speech, that is my right. Freedom of the press belongs to those who own one.

Usenet is not a public utility. Some Usenet sites are publicly funded or subsidized. Most of them, by plain count, are not. There is no government monopoly on Usenet, and little or no control.

Usenet is not a commercial network. Many Usenet sites are academic or government organizations; in fact, Usenet originated in academia. Therefore, there is a Usenet custom of keeping commercial traffic to a minimum. If such commercial traffic is generally considered worth carrying, then it may be grudgingly tolerated. Even so, it is usually separated somehow from non-commercial traffic; see comp.newprod.

Usenet is not the Internet. The Internet is a wide-ranging network, parts of which are subsidized by various governments. The Internet carries many kinds of traffic; Usenet is only one of them. And the Internet is only one of the various networks carrying Usenet traffic.

Usenet is not a Unix network, nor even an ASCII network.

Don't assume that everyone is using "rn" on a Unix machine. There are Vaxen running VMS, IBM mainframes, Amigas, and MS-DOS PCs reading

and posting to Usenet. And, yes, some of them use (shudder) EBCDIC. Ignore them if you like, but they're out there.

Usenet is not software. There are dozens of software packages used at various sites to transport and read Usenet articles. So no one program or package can be called "the Usenet software."

Software designed to support Usenet traffic can be (and is) used for other kinds of communication, usually without risk of mixing the two. Such private communication networks are typically kept distinct from Usenet by the invention of newsgroup names different from the universally-recognized ones.

Usenet is not a UUCP network.

UUCP is a protocol (some might say protocol suite, but that's a technical point) for sending data over point-to-point connections, typically using dialup modems. Usenet is only one of the various kinds of traffic carried via UUCP, and UUCP is only one of the various transports carrying Usenet traffic.

Well, enough negativity.

Propagation of News

In the old days, when UUCP over long-distance dialup lines was the dominant means of article transmission, a few well-connected sites had real influence in determining which newsgroups would be carried where. Those sites called themselves "the backbone."

But things have changed. Nowadays, even the smallest Internet site has connectivity the likes of which the backbone admin of yesteryear could only dream. In addition, in the U.S., the advent of cheaper long-distance

calls and high-speed modems has made long-distance Usenet feeds thinkable for smaller companies. There is only one pre-eminent UUCP transport site today in the U.S., namely UUNET. But UUNET isn't a player in the propagation wars, because it never refuses any traffic—it gets paid by the minute, after all; to refuse based on content would jeopardize its legal status as an enhanced service provider.

All of the above applies to the U.S. In Europe, different cost structures favored the creation of strictly controlled hierarchical organizations with central registries. This is all very unlike the traditional mode of U.S. sites (pick a name, get the software, get a feed, you're on). Europe's "benign monopolies", long uncontested, now face competition from looser organizations patterned after the U.S. model.

Group Creation

As discussed above, Usenet is not a democracy. Nevertheless, currently the most popular way to create a new newsgroup involves a "vote" to determine popular support for (and opposition to) a proposed newsgroup. Newsgroup Creation, for detailed instructions and guidelines on the process involved in making a newsgroup.

If you follow the guidelines, it is probable that your group will be created and will be widely propagated. However, due to the nature of Usenet, there is no way for any user to enforce the results of a newsgroup vote (or any other decision, for that matter). Therefore, for your new newsgroup to be propagated widely, you must not only follow the letter of the guidelines; you must also follow its spirit. And you must not allow even a whiff of shady dealings or dirty tricks to mar the vote.

So, you may ask: How is a new user supposed to know anything about the "spirit" of the guidelines? Obviously, she can't. This fact leads

inexorably to the following recommendation:

If you're a new user, don't try to create a new newsgroup alone.

If you have a good newsgroup idea, then read the news.groups newsgroup for a while (six months, at least) to find out how things work. If you're too impatient to wait six months, then you really need to learn; read news.groups for a year instead. If you just can't wait, find a Usenet old hand to run the vote for you.

Readers may think this advice unnecessarily strict. Ignore it at your peril. It is embarrassing to speak before learning. It is foolish to jump into a society you don't understand with your mouth open. And it is futile to try to force your will on people who can tune you out with the press of a key.

If You're Unhappy... Property rights being what they are, there is no higher authority on Usenet than the people who own the machines on which Usenet traffic is carried. If the owner of the machine you use says, "We will not carry alt.sex on this machine," and you are not happy with that order, you have no Usenet recourse. What can we outsiders do, after all?

That doesn't mean you are without options. Depending on the nature of your site, you may have some internal political recourse. Or you might find external pressure helpful. Or, with a minimal investment, you can get a feed of your own from somewhere else. Computers capable of taking Usenet feeds are down in the \$500 range now, Unix-capable boxes are going for under \$2000, and there are at least two Unix lookalikes in the \$100 price range.

No matter what, appealing to "Usenet" won't help. Even if those who read such an appeal regarding system administration are sympathetic to

your cause, they will almost certainly have even less influence at your site than you do.

By the same token, if you don't like what some user at another site is doing, only the administrator and/or owner of that site have any authority to do anything about it. Persuade them that the user in question is a problem for them, and they might do something (if they feel like it). If the user in question is the administrator or owner of the site from which he or she posts, forget it; you can't win. Arrange for your newsreading software to ignore articles from him or her if you can, and chalk one up to experience.

The History of Usenet (The ABCs)

In the beginning, there were conversations, and they were good. Then came Usenet in 1979, shortly after the release of V7 Unix with UUCP; and it was better. Two Duke University grad students in North Carolina, Tom Truscott and Jim Ellis, thought of hooking computers together to exchange information with the Unix community. Steve Bellovin, a grad student at the University of North Carolina, put together the first version of the news software using shell scripts and installed it on the first two sites: unc and duke. At the beginning of 1980 the network consisted of those two sites and phs (another machine at Duke), and was described at the January 1980 Usenix conference in Boulder, CO. {The Usenix conferences are semi-annual meetings where members of the Usenix Association, a group of Unix enthusiasts, meet and trade notes.} Steve Bellovin later rewrote the scripts into C programs, but they were never released beyond unc and duke. Shortly thereafter, Steve Daniel did another implementation in the C programming language for public distribution. Tom Truscott made further modifications, and this became the "A" news release.

In 1981 at the University of California at Berkeley, grad student Mark Horton and high school student Matt Glickman rewrote the news software

to add functionality and to cope with the ever increasing volume of news—"A" news was intended for only a few articles per group per day. This rewrite was the "B" news version. The first public release was version 2.1 in 1982; all versions before 2.1 were considered in beta test. As The Net grew, the news software was expanded and modified. The last version maintained and released primarily by Mark was 2.10.1.

Rick Adams, then at the Center for Seismic Studies, took over coordination of the maintenance and enhancement of the news software with the 2.10.2 release in 1984. By this time, the increasing volume of news was becoming a concern, and the mechanism for moderated groups was added to the software at 2.10.2. Moderated groups were inspired by ARPA mailing lists and experience with other bulletin board systems. In late 1986, version 2.11 of news was released, including a number of changes to support a new naming structure for newsgroups, enhanced batching and compression, enhanced ihave/sendme control messages, and other features. The current release of news is 2.11, patchlevel 19.

A new version of news, becoming known as "C" news, has been developed at the University of Toronto by Geoff Collyer and Henry Spencer. This version is a rewrite of the lowest levels of news to increase article processing speed, decrease article expiration processing and improve the reliability of the news system through better locking, etc. The package was released to The Net in the autumn of 1987. For more information, see the paper News Need Not Be Slow, published in the Winter 1987 Usenix Technical Conference proceedings.

Usenet software has also been ported to a number of platforms, from the Amiga and IBM PCs all the way to minicomputers and mainframes.

Hierarchies Newsgroups are organized according to their specific areas of concentration. Since the groups are in a tree structure, the various areas

are called hierarchies. There are seven major categories:

comp Topics of interest to both computer professionals and hobbyists, including topics in computer science, software sources, and information on hardware and software systems.

misc

Group addressing themes not easily classified into any of the other headings or which incorporate themes from multiple categories. Subjects include fitness, job-hunting, law, and investments.

sci Discussions marked by special knowledge relating to research in or application of the established sciences.

soc Groups primarily addressing social issues and socializing. Included are discussions related to many different world cultures.

talk Groups largely debate-oriented and tending to feature long discussions without resolution and without appreciable amounts of generally useful information.

news

Groups concerned with the news network, group maintenance, and software.

rec

Groups oriented towards hobbies and recreational activities

These "world" newsgroups are (usually) circulated around the entire Usenet—this implies world-wide distribution. Not all groups actually enjoy such wide distribution, however. The European Usenet and Eunet sites take only a selected subset of the more "technical" groups, and

controversial "noise" groups are often not carried by many sites in the U.S. and Canada (these groups are primarily under the talk and soc classifications). Many sites do not carry some or all of the comp.binaries groups because of the typically large size of the posts in them (being actual executable programs).

Also available are a number of "alternative" hierarchies:

alt True anarchy; anything and everything can and does appear; subjects include sex, the Simpsons, and privacy.

gnu

Groups concentrating on interests and software with the GNU Project of the Free Software Foundation. For further info on what the FSF is, FSF.

biz

Business-related groups.

Moderated vs Unmoderated

Some newsgroups insist that the discussion remain focused and on-target; to serve this need, moderated groups came to be. All articles posted to a moderated group get mailed to the group's moderator. He or she periodically (hopefully sooner than later) reviews the posts, and then either posts them individually to Usenet, or posts a composite digest of the articles for the past day or two. This is how many mailing list gateways work (for example, the Risks Digest).

news.groups & news.announce.newgroups

Being a good net.citizen includes being involved in the continuing growth and evolution of the Usenet system. One part of this involvement includes following the discussion in the groups news.groups and the notes in news.announce.newgroups. It is there that discussion goes on about the creation of new groups and destruction of inactive ones. Every person on Usenet is allowed and encouraged to vote on the creation of a newsgroup.

How Usenet Works

The transmission of Usenet news is entirely cooperative. Feeds are generally provided out of good will and the desire to distribute news everywhere. There are places which provide feeds for a fee (e.g. UUNET), but for the large part no exchange of money is involved.

There are two major transport methods, UUCP and NNTP. The first is mainly modem-based and involves the normal charges for telephone calls. The second, NNTP, is the primary method for distributing news over the Internet.

With UUCP, news is stored in batches on a site until the neighbor calls to receive the articles, or the feed site happens to call. A list of groups which the neighbor wishes to receive is maintained on the feed site. The Cnews system compresses its batches, which can dramatically reduce the transmission time necessary for a relatively heavy newsfeed.

NNTP, on the other hand, offers a little more latitude with how news is sent. The traditional store-and-forward method is, of course, available. Given the "real-time" nature of the Internet, though, other methods have been devised. Programs now keep constant connections with their news neighbors, sending news nearly instantaneously, and can handle dozens of simultaneous feeds, both incoming and outgoing.

The transmission of a Usenet article is centered around the unique Message-ID: header. When an NNTP site offers an article to a neighbor, it says it has that specific Message ID. If the neighbor finds it hasn't received the article yet, it tells the feed to send it through; this is repeated for each and every article that's waiting for the neighbor. Using unique IDs helps prevent a system from receiving five copies of an article from each of its five news neighbors, for example.

Further information on how Usenet works with relation to the various transports is available in the documentation for the Cnews and NNTP packages, as well as in RFC-1036, the Standard for Interchange of USENET Messages and RFC-977, Network News Transfer Protocol: A Proposed Standard for the Stream-Based Transmission of News. The RFCs do tend to be rather dry reading, particularly to the new user.

Mail Gateways

A natural progression is for Usenet news and electronic mailing lists to somehow become merged—which they have, in the form of news gateways. Many mailing lists are set up to "reflect" messages not only to the readership of the list, but also into a newsgroup. Likewise, posts to a newsgroup can be sent to the moderator of the mailing list, or to the entire mailing list. Some examples of this in action are comp.risks (the Risks Digest) and comp.dcom.telecom (the Telecom Digest).

This method of propagating mailing list traffic has helped solve the problem of a single message being delivered to a number of people at the same site—instead, anyone can just subscribe to the group. Also, mailing list maintenance is lowered substantially, since the moderators don't have to be constantly removing and adding users to and from the list. Instead, the people can read and not read the newsgroup at their leisure.

from "Dear Emily Postnews" by Brad Templeton Usenet "Netiquette"

There are many traditions with Usenet, not the least of which is dubbed netiquette—being polite and considerate of others. If you follow a few basic guidelines, you, and everyone that reads your posts, will be much happier in the long run.

Signatures

At the end of most articles is a small blurb called a person's signature. In Unix this file is named `.signature` in the person's login directory—it will vary for other operating systems. It exists to provide information about how to get in touch with the person posting the article, including their email address, phone number, address, or where they're located. Even so, signatures have become the graffiti of computers. People put song lyrics, pictures, philosophical quotes, even advertisements in their ".sigs". (Note, however, that advertising in your signature will more often than not get you flamed until you take it out.)

Four lines will suffice—more is just extra garbage for Usenet sites to carry along with your article, which is supposed to be the intended focus of the reader. Netiquette dictates limiting oneself to this "quota" of four—some people make signatures that are ten lines or even more, including elaborate ASCII drawings of their hand-written signature or faces or even the space shuttle. This is not cute, and will bother people to no end.

Similarly, it's not necessary to include your signature—if you forget to append it to an article, don't worry about it. The article's just as good as it ever would be, and contains everything you should want to say. Don't re-post the article just to include the signature.

Posting Personal Messages

If mail to a person doesn't make it through, avoid posting the message to a newsgroup. Even if the likelihood of that person reading the group is very high, all of the other people reading the articles don't give a whit what you have to say to Jim Morrison. Simply wait for the person to post again and double-check the address, or get in touch with your system administrator and see if it's a problem with local email delivery. It may also turn out that their site is down or is having problems, in which case it's just necessary to wait until things return to normal before contacting Jim.

Posting Mail

In the interests of privacy, it's considered extremely bad taste to post any email that someone may have sent, unless they explicitly give you permission to redistribute it. While the legal issues can be heavily debated, most everyone agrees that email should be treated as anything one would receive via normal snailmail, {The slang for the normal land and air postal service.} , with all of the assumed rights that are carried with it.

Test Messages

Many people, particularly new users, want to try out posting before actually taking part in discussions. Often the mechanics of getting messages out is the most difficult part of Usenet. To this end, many, many users find it necessary to post their tests to "normal" groups (for example, news.admin or comp.mail.misc). This is considered a major netiquette faux pas in the Usenet world. There are a number of groups available, called test groups, that exist solely for the purpose of trying out a news system, reader, or even new signature. They include

alt.test gnu.gnusenet.test misc.test

some of which will generate auto-magic replies to your posts to let you know they made it through. There are certain denizens of Usenet that frequent the test groups to help new users out. They respond to the posts, often including the article so the poster can see how it got to the person's site. Also, many regional hierarchies have test groups, like phl.test in Philadelphia.

By all means, experiment and test—just do it in its proper place.

Famous People Appearing

Every once in a while, someone says that a celebrity is accessible through "The Net"; or, even more entertaining, an article is forged to appear to be coming from that celebrity. One example is Stephen Spielberg—the rec.arts.movies readership was in an uproar for two weeks following a couple of posts supposedly made by Mr. Spielberg. (Some detective work revealed it to be a hoax.)

There are a few well-known people that are acquainted with Usenet and computers in general—but the overwhelming majority are just normal people. One should act with skepticism whenever a notable personality is "seen" in a newsgroup.

Summaries

Authors of articles occasionally say that readers should reply by mail and they'll summarize. Accordingly, readers should do just that—reply via mail. Responding with a followup article to such an article defeats the intention of the author. She, in a few days, will post one article containing the highlights of the responses she received. By following up to the whole group, the author may not read what you have to say.

When creating a summary of the replies to a post, try to make it as reader-friendly as possible. Avoid just putting all of the messages received into one big file. Rather, take some time and edit the messages into a form that contains the essential information that other readers would be interested in.

Also, sometimes people will respond but request to remain anonymous (one example is the employees of a corporation that feel the information's not proprietary, but at the same time want to protect themselves from political backlash). Summaries should honor this request accordingly by listing the From: address as anonymous or (Address withheld by request).

Quoting

When following up to an article, many newsreaders provide the facility to quote the original article with each line prefixed by > , as in

In article <1232@foo.bar.com>, sharon@foo.bar.com wrote: > I agree, I think that basketweaving's really catching on, > particularly in Pennsylvania. Here's a list of every person > in PA that currently engages in it publicly: line ... etc ...

This is a severe example (potentially a horribly long article), but proves a point. When you quote another person, edit out whatever isn't directly applicable to your reply. {But not changing their words, of course. } This gives the reader of the new article a better idea of what points you were addressing. By including the entire article, you'll only annoy those reading it. Also, signatures in the original aren't necessary; the readers already know who wrote it (by the attribution).

Avoid being tedious with responses—rather than pick apart an article, address it in parts or as a whole. Addressing practically each and every

word in an article only proves that the person responding has absolutely nothing better to do with his time.

If a "war" starts (insults and personal comments get thrown back and forth), take it into email—exchange email with the person you're arguing with. No one enjoys watching people bicker incessantly.

Crossposting

The Newsgroups: line isn't limited to just one group—an article can be posted in a list of groups. For instance, the line

Newsgroups: sci.space,comp.simulation

posts the article to both the groups sci.space and comp.simulation. It's usually safe to crosspost to up to three or four groups. To list more than that is considered "excessive noise."

It's also suggested that if an article is crossposted a Followup-To: header be included. It should name the group to which all additional discussion should be directed to. For the above example a possible Followup-To: would be

Followup-To: sci.space

which would make all followups automatically be posted to just sci.space, rather than both sci.space and comp.simulation. If every response made with a newsreader's "followup" command should go to the person posting the article no matter what, there's also a mechanism worked in to accommodate. The Followup-To: header should contain the single word poster:

Followup-To: poster

Certain newsreaders will use this to sense that a reply should never be posted back onto The Net. This is often used with questions that will yield a summary of information later, a vote, or an advertisement.

Recent News

One should avoid posting "recent" events—sports scores, a plane crash, or whatever people will see on the evening news or read in the morning paper. By the time the article has propagated across all of Usenet, the "news" value of the article will have become stale. (This is one case for the argument that Usenet news is a misnomer. {Note that the Clarinet News service (Clarinet) offers news items in a Usenet format as a precise alternative to the morning paper, et. al.})

Quality of Postings

How you write and present yourself in your articles is important. If you have terrible spelling, keep a dictionary near by. If you have trouble with grammar and punctuation, try to get a book on English grammar and composition (found in many bookstores and at garage sales). By all means pay attention to what you say—it makes you who you are on The Net.

Likewise, try to be clear in what you ask. Ambiguous or vague questions often lead to no response at all, leaving the poster discouraged. Give as much essential information as you feel is necessary to let people help you, but keep it within limits. For instance, you should probably include the operating system of your computer in the post if it's needed, but don't tell everybody what peripherals you have hanging off of it.

Useful Subjects

The Subject: line of an article is what will first attract people to read it— if it's vague or doesn't describe what's contained within, no one will read the article. At the same time, Subject: lines that're too wordy tend to be irritating. For example:

Good

Subject: Building Emacs on a Sun Sparc under 4.1

Good

Subject: Tryin' to find Waldo in NJ.

Bad

Subject: I can't get emacs to work !!!

Bad Subject: I'm desperately in search of the honorable Mr. Waldo in the state of...

Simply put, try to think of what will best help the reader when he or she encounters your article in a newsreading session.

Tone of Voice

Since common computers can't portray the inflection or tone in a person's voice, how articles are worded can directly affect the response to them. If you say

Anybody using a Vic-20 should go buy themselves a life.

you'll definitely get some responses—telling you to take a leap. Rather than be inflammatory, phrase your articles in a way that rationally expresses your opinion, like

What're the practical uses of a Vic-20 these days?

which presents yourself as a much more level-headed individual.

Also, what case (upper or lower) you use can indicate how you're trying to speak—netiquette dictates that if you USE ALL CAPITAL LETTERS, people will think you're "shouting." Write as you would in a normal letter to a friend, following traditional rules of English (or whatever language you happen to speak).

Computer Religion

No matter what kind of computer a person is using, theirs is always the best and most efficient of them all. Posting articles asking questions like What computer should I buy? An Atari ST or an Amiga? will lead only to fervent arguments over the merits and drawbacks of each brand. Don't even ask The Net—go to a local user group, or do some research of your own like reading some magazine reviews. Trying to say one computer is somehow better than another is a moot point.

The Anatomy of an Article

Frequently Asked Questions

A number of groups include Frequently Asked Question (FAQ) lists, which give the answers to questions or points that have been raised time and time again in a newsgroup. They're intended to help cut down on the redundant traffic in a group. For example, in the newsgroup alt.tv.simpsons, one recurring question is Did you notice that there's a different blackboard opening at the beginning of every Simpsons episode? As a result, it's part of the FAQ for that group.

Usually, FAQ lists are posted at the beginning of each month, and are set to expire one month later (when, supposedly, the next FAQ will be

published). Nearly every FAQ is also crossposted to news.answers, which is used as a Usenet repository for them.

The Pit-Manager Archive

MIT, with Jonathan Kamens, has graciously dedicated a machine to the archiving and storage of the various periodic postings that are peppered throughout the various Usenet groups. To access them, FTP to the system pit-manager.mit.edu and look in the directory /pub/usenet.

"Be it true or false, so it be news."

Ben Jonson, News from the New World

Telnet

Telnet is the main Internet protocol for creating a connection with a remote machine. It gives the user the opportunity to be on one computer system and do work on another, which may be across the street or thousands of miles away. Where modems are limited, in the majority, by the quality of telephone lines and a single connection, telnet provides a connection that's error-free and nearly always faster than the latest conventional modems.

Using Telnet

As with FTP (Anonymous FTP), the actual command for negotiating a telnet connection varies from system to system. The most common is telnet itself, though. It takes the form of:

```
telnet somewhere.domain
```

To be safe, we'll use your local system as a working example. By now, you hopefully know your site's domain name. If not, ask or try to figure it out. You'll not get by without it.

To open the connection, type

```
telnet your.system.name
```

If the system were wubba.cs.widener.edu, for example, the command would look like

```
telnet wubba.cs.widener.edu
```

The system will respond with something similar to

```
Trying 147.31.254.999...
Connected to wubba.cs.widener.edu.
Escape character is '^]'.
```

The escape character, in this example ^] (Control-]), is the character that will let you go back to the local system to close the connection, suspend it, etc. To close this connection, the user would type ^], and respond to the telnet> prompt with the command close. Local documentation should be checked for information on specific commands, functions, and escape character that can be used.

Telnet Ports

Many telnet clients also include a third option, the port on which the connection should take place. Normally, port 23 is the default telnet port; the user never has to think about it. But sometimes it's desirable to telnet to a different port on a system, where there may be a service available, or to aid in debugging a problem. Using

telnet somewhere.domain port

will connect the user to the given port on the system somewhere.domain. Many libraries use this port method to offer their facilities to the general Internet community; other services are also available. For instance, one would type

telnet martini.eecs.umich.edu 3000

to connect to the geographic server at the University of Michigan (Geographic Server). Other such port connections follow the same usage.

Publicly Accessible Libraries

Over the last several years, most university libraries have switched from a manual (card) catalog system to computerized library catalogs. The automated systems provide users with easily accessible and up-to-date information about the books available in these libraries. This has been further improved upon with the advent of local area networks, dialup modems, and wide area networks. Now many of us can check on our local library's holdings or that of a library halfway around the world!

Many, many institutions of higher learning have made their library catalogs available for searching by anyone on the Internet. They include Boston University, the Colorado Alliance of Research Libraries (CARL), and London University King's College.

To include a listing of some of the existing sites would not only be far too long for this document, it would soon be out of date. Instead, several lists are being maintained and are available either by mail or via FTP. Also, the Internet Resource Guide (IRG) also describes a few libraries that are accessible—IRG for further information.

Art St. George and Ron Larsen are maintaining a list of Internet-accessible libraries and databases often referred to as "the St. George directory." It began with only library catalogs but has expanded to include sections on campus-wide information systems, and even bulletin board systems that are not on the Internet. The library catalog sections are divided into those that are free, those that charge, and international (i.e. non-U.S.) catalogs; they are arranged by state, province, or country within each section. There is also a section giving dialup information for some of the library catalogs. It's available for FTP (Anonymous FTP) on nic.cerf.net in the directory cerfnet/cerfnet_info/library_catalog. The file internet-catalogs has a date suffix; check for the most current date. The information is updated periodically.

Billy Barron, Systems Manager at the University of North Texas, produces a directory as an aid to his user community. It complements the St. George guide by providing a standard format for all systems which lists the Internet address, login instructions, the system vendor, and logoff information. The arrangement is alphabetic by organization name. It's available for FTP on vaxb.acs.unt.edu in the subdirectory library as the file libraries.txt.

For announcements of new libraries being available and discussion on related topics, consult the Usenet newsgroup comp.internet.library (Usenet News to learn how to read news).

Bulletin Board Systems

The Cleveland Freenet

Freenets are open-access, free, community computer systems. One such system is the Cleveland Freenet, sponsored by CWRU (Case Western Reserve University). Anyone and everyone is welcome to join and take part

in the exciting project—that of a National Telecomputing Public Network, where everyone benefits. There's no charge for the registration process and no charge to use the system.

To register, telnet to any one of

freenet-in-a.cwru.edu freenet-in-b.cwru.edu freenet-in-c.cwru.edu

After you're connected, choose the entry on the menu that signifies you're a guest user. Another menu will follow; select Apply for an account, and you'll be well on your way to being a FreeNet member.

You will need to fill out a form and send it to them through the Postal Service—your login id and password will be created in a few days. At that point you're free to use the system as you wish. They provide multi-user chat, email, Usenet news, and a variety of other things to keep you occupied for hours on end.

Directories

There are a few systems that are maintained to provide the Internet community with access to lists of information—users, organizations, etc. They range from fully dedicated computers with access to papers and research results, to a system to find out about the faculty members of a university.

Knowbot

Knowbot is a "master directory" that contains email address information from the NIC WHOIS database (Whois), the PSI White Pages Pilot Project, the NYSERNET X.500 database and MCI Mail. Most of these services are email registries themselves, but Knowbot provides a very comfortable way to access all of them in one place. Telnet to nri.reston.va.us on port 185.

White Pages

PSI maintains a directory of information on individuals. It will list the person's name, organization, and email address if it is given. Telnet to wp.psi.net and log in as fred. The White Pages Project also includes an interface to use Xwindows remotely.

Faculty and Staff Listings

Many universities offer access to information on current faculty and staff. Included are:

Cornell Telnet to cuinfo.cornell.edu on port 3000.
NC State Telnet to ccvax1.cc.ncsu.edu and log in as info.
Rutgers Telnet to hangout.rutgers.edu on port 98.
U of Maryland Telnet to umail.umd.edu and log in as lookup.
UNC Chapel Hill Telnet to info.acs.unc.edu and log in as info.
Yale Telnet to yalevm.ycc.yale.edu on port 300.

Databases

For information on database services, Commercial Databases. Not all databases on the Internet require payment for use, though. There do exist some, largely research-driven databases, which are publicly accessible. New ones spring up regularly.

To find out more about the databases in this section, contact the people directly responsible for them. Their areas of concentration and the software used to implement them are widely disparate, and are probably beyond the author's expertise. Also, don't forget to check with your local library—the reference librarian there can provide information on conventional resources,

and possibly even those available over the Internet (they are becoming more common).

Colorado Alliance of Research Libraries (CARL)

The Colorado Alliance of Research Libraries (CARL), in association with CARL Systems Inc., operates a public access catalog of services. Offered are a number of library databases, including searches for government periodicals, book reviews, indices for current articles, and access to to other library databases around the country. Other services are available to CARL members including an online encyclopedia. Telnet to [pac.carl.org](mailto:help@carl.org), or write to help@carl.org for more details.

PENpages

PENpages is an agriculturally-oriented database administered by Pennsylvania State University. Information entered into PENpages is provided by numerous sources including the Pennsylvania Dept. of Agriculture, Rutgers University, and Penn State. Easy-to-use menus guide users to information ranging from cattle and agricultural prices to current weather information, from health information to agricultural news from around the nation. A keyword search option also allows users to search the database for related information and articles. The database is updated daily, and a listing of most recent additions is displayed after login. Telnet to psupen.psu.edu and log in as the user PNOTPA.

Clemson Univ. Forestry & Agricultural Network

Clemson maintains a database similar to PENpages in content, but the information provided tends to be localized to the Southeastern United States. A menu-driven database offers queries involving the weather, food, family, and human resources. Telnet to eureka.clemson.edu and log in as

PUBLIC. You need to be on a good VT100 emulator (or a real VT terminal).

University of Maryland Info Database

The Computer Science department of the University of Maryland maintains a repository of information on a wide variety of topics. They wish to give a working example of how network technology can (and should) provide as much information as possible to those who use it. Telnet to info.umd.edu and log in as info. The information contained in the database is accessible through a screen-oriented interface, and everything therein is available via anonymous FTP.

There is a mailing list used to discuss the UMD Info Database, welcoming suggestions for new information, comments on the interface the system provides, and other related topics. Send mail to listserv@umdd.umd.edu with a body of

subscribe INFO-L Your Full Name

Listservs for more information on using the Listserv system.

University of Michigan Weather Underground

The University of Michigan's Department of Atmospheric, Oceanic, & Space Sciences maintains a database of weather and related information for the United States and Canada. Available are current weather conditions and forecasts for cities in the U.S., a national weather summary, ski conditions, earthquake and hurricane updates, and a listing of severe weather conditions. Telnet to madlab.spri.umich.edu on port 3000 to use the system.

Geographic Name Server

A geographic database listing information for cities in the United States and some international locations is maintained by Merit, Inc. The database is searchable by city name, zip code, etc. It will respond with a lot of information: the area code, elevation, time zone, and longitude and latitude are included. For example, a query of 19013 yields

```
0 Chester 1 42045 Delaware 2 PA Pennsylvania 3 US United States F 45
Populated place L 39 50 58 N 75 21 22 W P 45794 E 22 Z 19013 Z 19014 Z
19015 Z 19016 ..
```

To use the server, telnet to martini.eecs.umich.edu on port 3000. The command help will yield further instructions, along with an explanation for each of the fields in a response.

FEDIX—Minority Scholarship Information

FEDIX is an on-line information service that links the higher education community and the federal government to facilitate research, education, and services. The system provides accurate and timely federal agency information to colleges, universities, and other research organizations. There are no registration fees and no access charges for FEDIX whatsoever.

FEDIX offers the Minority On-Line Information Service (MOLIS), a database listing current information about Black and Hispanic colleges and universities.

Daily information updates are made on federal education and research programs, scholarships, fellowships, and grants, available used research equipment, and general information about FEDIX itself. To access the database, telnet to fedix.fie.com and log in as fedix.

Science & Technology Information System

The STIS is maintained by the National Science Foundation (NSF), and provides access to many NSF publications. The full text of publications can be searched online and copied from the system, which can accommodate up to ten users at one time. Telnet to stis.nsf.gov and log in as public. Everything on the system is also available via anonymous FTP. For further information, contact:

STIS, Office of Information Systems, Room 401 National Science Foundation 1800 G. Street, N.W. Washington, D.C. 20550 stis-request@nsf.gov (202) 357-7492 (202) 357-7663 (Fax)

Ocean Network Information Center

The University of Delaware College of Marine Studies offers access to an interactive database of research information covering all aspects of marine studies, nicknamed OCEANIC. This includes the World Oceanic Circulation Experiment (WOCE) information and program information, research ship schedules and information, and a Who's Who of email and mailing addresses for oceanic studies. Data from a variety of academic institutions based on research studies is also available. Telnet to delocn.udel.edu and log in as INFO.

NASA/IPAC Extragalactic Database (NED)

The NASA/IPAC Extragalactic Database (NED) is an ongoing project, funded by NASA, to make data and literature on extragalactic objects available over computer networks. NED is an object-oriented database which contains extensive information for nearly 132,000 extragalactic objects taken from about major catalogs of galaxies, quasars, infrared and radio sources. NED provides positions, names, and other basic data (e.g. magnitude types, sizes and redshifts as well as bibliographic references and abstracts). Searches can be done by name, around a name, and on an

astronomical position. NED contains a tutorial which guides the user through the retrieval process. Telnet to ipac.caltech.edu and log in as ned.

U.S. Naval Observatory Automated Data Service

Operated by the U.S. Naval Observatory in Washington, D.C., this automated data service provides database access to information ranging from current navigational satellite positioning, astronomical data, and software utilities. A wide variety of databases can be searched and instructions for file transfer are given. Telnet to tycho.usno.navy.mil and log in as ads.

"My consciousness suddenly switched locations, for the first time in my life, from the vicinity of my head and body to a point about twenty feet away from where I normally see the world." Howard Rheingold, *Virtual Reality* p255

Various Tools

New and interesting ways to use the Internet are being dreamed up every day. As they gain wide-spread use, some methods become near-standard (or actual written standard) tools for Internet users to take advantage of. A few are detailed here; there are undoubtedly others, and new ideas spring up all the time. An active user of the Internet will discover most of the more common ones in time. Usually, these services are free. Commercial Services for applications that are commercially available over the Internet.

Usenet is often used to announce a new service or capability on the Internet. In particular, the groups comp.archives and comp.protocols.tcp-ip are good places to look. Information will drift into other areas as word spreads. Usenet News for information on reading news.

Finger

On many systems there exists the finger command, which yield information about each user that's currently logged in. This command also has extensions for use over the Internet, as well. Under normal circumstances, the command is simply finger for a summary of who's logged into the local system, or finger username for specific information about a user. It's also possible to go one step further and go onto the network. The general usage is

```
finger @hostname
```

To see who's currently logged in at Widener University, for instance, use

```
% finger @cs.widener.edu [cs.widener.edu] Login Name TTY Idle When  
Where brendan Brendan Kehoe p0 Fri 02:14 tattoo.cs.widene sven Sven  
Heinicke p1 Fri 04:16 xyplex3.cs.widen
```

To find out about a certain user, they can be fingered specifically (and need not be logged in):

```
% finger bart@cs.widener.edu  
[cs.widener.edu]  
Login name: bart In real life: Bart Simpson  
Directory: /home/springfield/bart Shell: /bin/underachiever  
Affiliation: Brother of Lisa Home System: channel29.fox.org  
Last login Thu May 23 12:14 (EDT) on tty6 from channel29.fox.org.  
No unread mail  
Project: To become a "fluff" cartoon character.  
Plan:  
Don't have a cow, man.
```

Please realize that some sites are very security conscious, and need to restrict the information about their systems and users available to the outside world. To that end, they often block finger requests from outside sites—so don't be surprised if fingering a computer or a user returns with Connection refused.

Internet Relay Chat

The Lamont View Server System

On lamont.ldgo.columbia.edu in pub/gb.tar.Z.

Ping

The ping command allows the user to check if another system is currently "up" and running. The general form of the command is ping system. {The usage will, again, vary.} For example,

```
ping cs.widener.edu
```

will tell you if the main machine in Widener University's Computer Science lab is currently online (we certainly hope so!).

Many implementations of ping also include an option to let you see how fast a link is running (to give you some idea of the load on the network). For example:

```
% ping -s cs.swarthmore.edu PING cs.swarthmore.edu: 56 data bytes 64
bytes from 130.58.68.1: icmp_seq=0 ttl=251 time=66 ms 64 bytes from
130.58.68.1: icmp_seq=1 ttl=251 time=45 ms 64 bytes from 130.58.68.1:
icmp_seq=2 ttl=251 time=46 ms ^C --- cs.swarthmore.edu ping statistics
--- 3 packets transmitted, 3 packets received, 0% packet loss round-trip
min/avg/max = 45/52/66 ms
```

This case tells us that for cs.swarthmore.edu it takes about 46 milliseconds for a packet to go from Widener to Swarthmore College and back again. It also gives the average and worst-case speeds, and any packet loss that may have occurred (e.g. because of network congestion).

While ping generally doesn't hurt network performance, you shouldn't use it too often—usually once or twice will leave you relatively sure of the other system's state.

Talk

Sometimes email is clumsy and difficult to manage when one really needs to have an interactive conversation. The Internet provides for that as well, in the form of talk. Two users can literally see each other type across thousands of miles.

To talk with Bart Simpson at Widener, one would type

```
talk bart@@cs.widener.edu
```

which would cause a message similar to the following to be displayed on Bart's terminal:

```
Message from Talk_Daemon@cs.widener.edu at 21:45 ... talk:
connection requested by joe@ee.someplace.edu talk: respond with: talk
joe@ee.someplace.edu
```

Bart would, presumably, respond by typing `talk joe@ee.someplace.edu`. They could then chat about whatever they wished, with instantaneous response time, rather than the write-and-wait style of email. To leave talk, on many systems one would type Ctrl-C (hold down the Control key and press C). Check local documentation to be sure.

There are two different versions of talk in common use today. The first, dubbed "old talk," is supported by a set of Unix systems (most notably, those currently sold by Sun). The second, ntalk (aka "new talk"), is more of the standard. If, when attempting to talk with another user, it responds with an error about protocol families, odds are the incompatibilities between versions of talk is the culprit. It's up to the system administrators of sites which use the old talk to install ntalk for their users.

Wide Area Information Servers (WAIS)

The WHOIS Database

The main WHOIS database is run at the Network Information Center (NIC). The whois command will let you search a database of every registered domain (e.g. mit.edu) and of registered users. It's primarily used by system postmasters or listowners to find the Points of Contact for a site, to let them know of a problem or contact them for one reason or another. You can also find out their postal address. For example:

```
% whois mit.edu
```

```
Massachusetts Institute of Technology (MIT) MIT.EDU 18.72.2.1
```

```
Massachusetts Institute of Technology (MIT-DOM) MIT.EDU
```

Note that there are two entries for mit.edu; we'll go for the second.

```
% whois mit-dom
```

```
Massachusetts Institute of Technology (MIT-DOM)
```

```
Cambridge, MA 02139
```

```
Domain Name: MIT.EDU
```

```
Administrative Contact, Technical Contact, Zone Contact: Schiller,  
Jeffrey I. (JIS) JIS@MIT.EDU (617) 253-8400
```

Record last updated on 22-Jun-88.

Domain servers in listed order:

STRAWB.MIT.EDU 18.71.0.151

W20NS.MIT.EDU 18.70.0.160

BITSY.MIT.EDU 18.72.0.3

LITHIUM.LCS.MIT.EDU 18.26.0.121

To see this host record with registered users, repeat the command with a star (*) before the name; or, use '%' to show JUST the registered users.

Much better! Now this information (sought, possibly, by a system administrator) can be used to find out how to notify MIT of a security issue or problem with connectivity.

Queries can be made for individuals as well; the following would yield an entry for the author:

% whois brendan

Kehoe, Brendan (BK59) brendan@cs.widener.edu

Widener University

Department of Computer Science

Kirkbride 219

P.O. Box 83 Widener University

Chester, PA 19013

(215)/499-4011

Record last updated on 02-May-91.

Included is the author's name, his handle (a unique sequence of letters and numbers), information on how to contact him, and the last time the record was modified in any way.

Anyone can register with the whois database. People who are administrative or technical contacts for domains are registered automatically when their domain applications are processed. For normal users, one must simply fill out a form from the NIC. FTP to nic.ddn.mil and get the file netinfo/user-template.txt. The completed form should be mailed to registrar@nic.ddn.mil.

Other Uses of WHOIS

Also, many educational sites run WHOIS servers of their own, to offer information about people who may be currently on the staff or attending the institution. To specify a WHOIS server, many implementations include some sort of option or qualifier—in VMS under MultiNet, it's /HOST, in Unix -h. To receive information about using the Stanford server, one might use the command

```
whois -h stanford.edu help
```

A large list of systems offering WHOIS services is being maintained by Matt Power of MIT (mhpower@stan.mit.edu). It is available via anonymous FTP from sipb.mit.edu, in the directory pub/whois. The file is named whois-servers.list.

The systems available include, but are certainly not limited to, Syracuse University (syr.edu), New York University (acfcluster.nyu.edu), the University of California at San Diego (ucsd.edu), and Stanford University (stanford.edu).

"Fingers were made before forks."
Jonathan Swift, Polite Conversation

Commercial Services

Many services can be accessed through the Internet. As time progresses and more outlets for commercial activity appear, once-restricted traffic (by the NSFnet Acceptable Use Policy) may now flow freely. Now that there are other networks for that information to travel on, businesses are making their move.

Internet Service Providers

Providers (AlterNet, PSI, etc)...

Supercomputers The Internet Resource Guide (IRG) contains a chapter on computer time that's available for a fee. Rather than reproduce it here, which would fast become out-of-date as well as triple the size of this guide, it's suggested that the reader consult the IRG if such services are of interest.

Electronic Journals

The Association of Research Libraries (ARL) publishes a hard-copy directory of electronic journals, newsletters, and scholarly discussion lists. It is a compilation of entries for hundreds of

sts, dozens of journals and newsletters, and a many "other" titles, including newsletter-digests, into one reference source. Each entry includes instructions on how to access the referenced publication or list.

The documents are available electronically by sending the commands

```
get ejournl1 directry get ejournl2 directry
```

to the server at LISTSERV@OTTAWA.BITNET. Listservs for further instructions on using a listserv.

The directory, along with a compilation by Diane Kovacs called Directories of Academic E-Mail Conferences, is available in print and on diskette (DOS WordPerfect and MacWord) from:

Office of Scientific & Academic Publishing Association of Research Libraries
1527 New Hampshire Avenue, NW Washington, DC 20036
ARLHQ@UMDC.BITNET (202) 232—2466 (202) 462—7849 (Fax)

The ARL is a not-for-profit organization representing over one hundred research libraries in the United States and Canada. The publication is available to ARL members for \$10 and to non-members for \$20 (add \$5 postage per directory for foreign addresses). Orders of six or more copies will receive a 10% discount; all orders must be prepaid and sent to the ARL.

Commercial Databases

The American Institute of Physics maintains the Physics Information Network. It contains the bibliographic SPIN and General Physics Advanced Abstracts databases. Also available is access to bulletin boards and several searchable lists (job notices, announcements, etc). Telnet to pinet.aip.org; new users must log in as NEW and give registration information.

Some of the databases accessible through WAIS (WAIS) are available for a fee.

Clarinet News

Clarinet's an electronic publishing network service that provides professional news and information, including live UPI wireservice news, in the Usenet file format.

Clarinet lets you read an "electronic newspaper" right on the local system; you can get timely industry news, technology related wirestories, syndicated columns and features, financial information, stock quotes and more.

Clarinet's provided by using the Usenet message interchange format, and is available via UUCP and other delivery protocols, including NNTP.

The main feature is ClariNews, an "electronic newspaper," gathered live from the wire services of United Press International (UPI). ClariNews articles are distributed in 100 newsgroups based on their subject matter, and are keyworded for additional topics and the geographical location of the story. ClariNews includes headlines, industry news, box scores, network TV schedules, and more. The main products of ClariNews are:

ClariNews General, the general news"paper" with news, sports, and features, averaging about 400 stories per day.

TechWire, special groups for stories on science, technology, and industry stories around them.

ClariNews-Biz, business and financial stories.

Newsbytes, a daily computer industry newsmagazine.

Syndicated Columns, including Dave Barry (humor) and Mike Royko (opinion).

Full information on ClariNet, including subscription information, is available from

Clarinet Communications Corp. 124 King St. North Waterloo, Ontario
N2J 2X8 info@@clarinet.com (800) USE-NETS

or with anonymous FTP in the directory /Clarinet on ftp.uu.net
(Anonymous FTP).

"Needless to say, Aristotle did not envisage modern finance."
Frederick Copleston, S.J.
A History of Philosophy: Vol 1 Greece & Rome Part II, p95

Things You'll Hear About

There are certain things that you'll hear about shortly after you start actively using the Internet. Most people assume that everyone's familiar with them, and they require no additional explanation. If only that were true!

This section addresses a few topics that are commonly encountered and asked about as a new user explores Cyberspace. Some of them are directly related to how the networks are run today; other points are simply interesting to read about.

The Internet Worm

from a letter by Severo M. Ornstein, in ACM June 89 Vol32 No6 and the appeal notice

On November 2, 1988, Robert Morris, Jr., a graduate student in Computer Science at Cornell, wrote an experimental, self-replicating, self-propagating program called a worm and injected it into the Internet. He chose to release it from MIT, to disguise the fact that the worm came from Cornell. Morris soon discovered that the program was replicating and

reinfecting machines at a much faster rate than he had anticipated—there was a bug. Ultimately, many machines at locations around the country either crashed or became "catatonic." When Morris realized what was happening, he contacted a friend at Harvard to discuss a solution. Eventually, they sent an anonymous message from Harvard over the network, instructing programmers how to kill the worm and prevent reinfection. However, because the network route was clogged, this message did not get through until it was too late. Computers were affected at many sites, including universities, military sites, and medical research facilities. The estimated cost of dealing with the worm at each installation ranged from \$200 to more than \$53,000. {Derived in part from a letter by Severo M. Ornstein, in the Communications of the ACM, Vol 32 No 6, June 1989.}

The program took advantage of a hole in the debug mode of the Unix sendmail program, which runs on a system and waits for other systems to connect to it and give it email, and a hole in the finger daemon fingerd, which serves finger requests (Finger). People at the University of California at Berkeley and MIT had copies of the program and were actively disassembling it (returning the program back into its source form) to try to figure out how it worked.

Teams of programmers worked non-stop to come up with at least a temporary fix, to prevent the continued spread of the worm. After about twelve hours, the team at Berkeley came up with steps that would help retard the spread of the virus. Another method was also discovered at Purdue and widely published. The information didn't get out as quickly as it could have, however, since so many sites had completely disconnected themselves from the network.

After a few days, things slowly began to return to normalcy and everyone wanted to know who had done it all. Morris was later named in The New

York Times as the author (though this hadn't yet been officially proven, there was a substantial body of evidence pointing to Morris).

Robert T. Morris was convicted of violating the computer Fraud and Abuse Act (Title 18), and sentenced to three years of probation, 400 hours of community service, a fine of \$10,050, and the costs of his supervision. His appeal, filed in December, 1990, was rejected the following March.

The Cuckoo's Egg

First in an article entitled "Stalking the Wily Hacker," and later in the book *The Cuckoo's Egg*, Clifford Stoll detailed his experiences trying to track down someone breaking into a system at Lawrence Berkeley Laboratory in California. {See the bibliography for full citations.}

A 75-cent discrepancy in the Lab's accounting records led Stoll on a chase through California, Virginia, and Europe to end up in a small apartment in Hannover, West Germany. Stoll dealt with many levels of bureaucracy and red tape, and worked with the FBI, the CIA, and the German Bundespost trying to track his hacker down.

The experiences of Stoll, and particularly his message in speaking engagements, have all pointed out the dire need for communication between parties on a network of networks. The only way everyone can peacefully co-exist in Cyberspace is by ensuring rapid recognition of any existing problems.

Organizations

The indomitable need for humans to congregate and share their common interests is also present in the computing world. User groups exist around the world, where people share ideas and experiences. Similarly, there are

organizations which are one step "above" user groups; that is to say, they exist to encourage or promote an idea or set of ideas, rather than support a specific computer or application of computers.

The Association for Computing Machinery

The Association for Computing Machinery (the ACM) was founded in 1947, immediately after Eckert and Mauchly unveiled one of the first electronic computers, the ENIAC, in 1946. Since then, the ACM has grown by leaps and bounds, becoming one of the leading educational and scientific societies in the computer industry.

The ACM's stated purposes are:

To advance the sciences and arts of information processing;

To promote the free interchange of information about the sciences and arts of information processing both among specialists and among the public;

To develop and maintain the integrity and competence of individuals engaged in the practices of the sciences and arts of information processing.

Membership in the ACM has grown from seventy-eight in September, 1947, to over 77,000 today. There are local chapters around the world, and many colleges and universities endorse student chapters. Lecturers frequent these meetings, which tend to be one step above the normal "user group" gathering. A large variety of published material is also available at discounted prices for members of the association.

The ACM has a number of Special Interest Groups (SIGs) that concentrate on a certain area of computing, ranging from graphics to the Ada programming language to security. Each of the SIGs also publishes its

own newsletter. There is a Usenet group, comp.org.acm, for the discussion of ACM topics. Usenet News for more information on reading news.

For more information and a membership application, write to:

Association for Computing Machinery 1515 Broadway New York City,
NY 10036 ACMHELP@ACMVM.BITNET (212) 869-7440

Computer Professionals for Social Responsibility from their letter to prospective members

The CPSR is an alliance of computer professionals concentrating on certain areas of the impact of computer technology on society. It traces its history to the fall of 1981, when several researchers in Palo Alto, California, organized a lunch meeting to discuss their shared concerns about the connection between computing and the nuclear arms race. Out of that meeting and the discussions which followed, CPSR was born, and has been active ever since. {This section is part of the CPSR's letter to prospective members.}

The national CPSR program focuses on the following project areas:

Reliability and Risk This area reflects on the concern that overreliance on computing technology can lead to unacceptable risks to society. It includes, but isn't limited to, work in analyzing military systems such as SDI.

Civil Liberties and Privacy This project is concerned with such topics as the FBI National Crime Information Center, the growing use of databases by both government and private industry, the right of access to public information, extension of First Amendment rights to electronic communication, and establishing legal protections for privacy of computerized information.

Computers in the Workplace The CPSR Workplace Project has concentrated its attention on the design of software for the workplace, and particularly on the philosophy of "participatory design," in which software designers work together with users to ensure that systems meet the actual needs of that workplace.

The 21st Century Project This is a coalition with other professional organizations working towards redirecting national research priorities from concentrating on military issues to anticipating and dealing with future problems as science and technology enter the next century.

For more information on the CPSR, contact them at:

Computer Professionals for Social Responsibility P.O. Box 717 Palo Alto, CA 94302 cpsr@csli.stanford.edu (415) 322—3778 (415) 322—3798 (Fax)

The Electronic Frontier Foundation

The Electronic Frontier Foundation (EFF) was established to help civilize the "electronic frontier"—the Cyberspatial medium becoming ever-present in today's society; to make it truly useful and beneficial not just to a technical elite, but to everyone; and to do this in a way which is in keeping with the society's highest traditions of the free and open flow of information and communication. {This section was derived from eff.about, available along with other material via anonymous FTP from ftp.eff.org}

The mission of the EFF is to engage in and support educational activities which increase popular understanding of the opportunities and challenges posed by developments in computing and telecommunications;

to develop among policy-makers a better understanding of the issues underlying free and open telecommunications, and support the creation of legal and structural approaches which will ease the assimilation of these new technologies by society;

to raise public awareness about civil liberties issues arising from the rapid advancement in the area of new computer-based communications media and, where necessary, support litigation in the public interest to preserve, protect, and extend First Amendment rights within the realm of computing and telecommunications technology;

to encourage and support the development of new tools which will endow non-technical users with full and easy access to computer-based telecommunications;

The Usenet newsgroups `comp.org.eff.talk` and `comp.org.eff.news` are dedicated to discussion concerning the EFF. They also have mailing list counterparts for those that don't have access to Usenet, `eff-talk-request@eff.org` and `eff-news-request@eff.org`. The first is an informal arena (aka a normal newsgroup) where anyone may voice his or her opinions. The second, `comp.org.eff.news`, is a moderated area for regular postings from the EFF in the form of EFFector Online. To submit a posting for the EFFector Online, or to get general information about the EFF, write to `eff@eff.org`. There is also a wealth of information available via anonymous FTP on `ftp.eff.org`.

The EFF can be contacted at

The Electronic Frontier Foundation, Inc. 155 Second St. #1 Cambridge, MA 02141 `eff@eff.org` (617) 864-0665 (617) 864-0866 (Fax)

The Free Software Foundation

The Free Software Foundation was started by Richard Stallman (creator of the popular GNU Emacs editor). It is dedicated to eliminating restrictions on copying, redistributing, and modifying software.

The word "free" in their name does not refer to price; it refers to freedom. First, the freedom to copy a program and redistribute it to your neighbors, so that they can use it as well as you. Second, the freedom to change a program, so that you can control it instead of it controlling you; for this, the source code must be made available to you.

The Foundation works to provide these freedoms by developing free compatible replacements for proprietary software. Specifically, they are putting together a complete, integrated software system called "GNU" that is upward-compatible with Unix. {As an aside, the editor of the GNU project, emacs, contains a built-in LISP interpreter and a large part of its functionality is written in LISP. The name GNU is itself recursive (the mainstay of the LISP language); it stands for "Gnu's Not Unix."}

When it is released, everyone will be permitted to copy it and distribute it to others. In addition, it will be distributed with source code, so you will be able to learn about operating systems by reading it, to port it to your own machine, and to exchange the changes with others.

For more information on the Free Software Foundation and the status of the GNU Project, or for a list of the current tasks that still need to be done, write to gnu@prep.ai.mit.edu.

The IEEE

Need IEEE...

The League for Programming Freedom

The League for Programming Freedom is a grass-roots organization of professors, students, businessmen, programmers and users dedicated to "bringing back" the freedom to write programs, which they contend has been lost over the past number years. The League is not opposed to the legal system that Congress intended—copyright on individual programs. Their aim is to reverse the recent changes made by judges in response to special interests, often explicitly rejecting the public interest principles of the Constitution.

The League works to abolish the new monopolies by publishing articles, talking with public officials, boycotting egregious offenders, and in the future may intervene in court cases. On May 24, 1989, the League picketed Lotus headquarters because of their lawsuits, and then again on August 2, 1990. These marches stimulated widespread media coverage for the issue. They welcome suggestions for other activities, as well as help in carrying them out.

For information on the League and how to join, write to

League for Programming Freedom 1 Kendall Square #143 P.O. Box 9171
Cambridge, MA 02139 league@prep.ai.mit.edu

Networking Initiatives

Research and development are two buzz words often heard when discussing the networking field—everything needs to go faster, over longer distances, for a lower cost. To "keep current," one should read the various trade magazines and newspapers, or frequent the networking-oriented newsgroups of Usenet. If possible, attend trade shows and symposia like Usenix, Interop, et. al.

ISDN

NREN

The National Research and Education Network (NREN) is a five-year project approved by Congress in the Fall of 1991. It's intended to create a national electronic "super-highway." The NREN will be 50 times faster than the fastest available networks (at the time of this writing). Proponents of the NREN claim it will be possible to transfer the equivalent of the entire text of the Encyclopedia Britannica in one second. Further information, including the original text of the bill presented by Senator Al Gore (D—TN), is available through anonymous FTP to [nis.nsf.net](ftp://nis.nsf.net), in the directory `nsfnet`. In addition, Vint Cerf wrote on the then-proposed NREN in RFC-1167, *Thoughts on the National Research and Education Network*. RFCs for information on obtaining RFCs.

A mailing list, nren-discuss@uu.psi.com, is available for discussion of the NREN; write to nren-discuss-request@uu.psi.com to be added.

"To talk in publick, to think in solitude, to read and to hear, to inquire, and to answer inquiries, is the business of a scholar." Samuel Johnson
Chapter VIII The History of Rasselas, Prince of Abissinia

——— Finding Out More

Internet Resource Guide

The NSF Network Service Center (NNSC) compiles and makes available an Internet Resource Guide (IRG). The goal of the guide is to increase the visibility of various Internet resources that may help users do their work better. While not yet an exhaustive list, the guide is a useful compendium of many resources and can be a helpful reference for a new user.

Resources listed are grouped by types into sections. Current sections include descriptions of online library catalogs, data archives, online white

pages directory services, networks, network information centers, and computational resources, such as supercomputers. Each entry describes the resource, identifies who can use the resource, explains how to reach the local network via the Internet, and lists contacts for more information. The list is distributed electronically by the NNSC. To receive a guide, or to get on a mailing list that alerts you to when it is updated, send a message to resource-guide-request@nnsf.net.

The current edition of the IRG is available via anonymous FTP from nnsf.net, in the directory `/resource-guide`.

Requests for Comments

The internal workings of the Internet are defined by a set of documents called RFCs (Request for Comments). The general process for creating an RFC is for someone wanting something formalized to write a document describing the issue and mailing it to Jon Postel (postel@isi.edu). He acts as a referee for the proposal. It is then commented upon by all those wishing to take part in the discussion (electronically, of course). It may go through multiple revisions. Should it be generally accepted as a good idea, it will be assigned a number and filed with the RFCs.

The RFCs can be divided into five groups: required, suggested, directional, informational and obsolete. Required RFCs (e.g., RFC-791, The Internet Protocol) must be implemented on any host connected to the Internet.

Suggested RFCs are generally implemented by network hosts. Lack of them does not preclude access to the Internet, but may impact its usability. RFC-793, Transmission Control Protocol, is a must for those implementing TCP.

Directional RFCs were discussed and agreed to, but their application has never come into wide use. This may be due to the lack of wide need for the specific application (RFC-937, The Post Office Protocol) or that, although technically superior, ran against other pervasive approaches (RFC-891, Hello). It is suggested that, should the facility be required by a particular site, an implementation be done in accordance with the RFC. This ensures that, should the idea be one whose time has come, the implementation will be in accordance with some standard and will be generally usable.

Informational RFCs contain factual information about the Internet and its operation (RFC-990, Assigned Numbers).

There is also a subset of RFCs called FYIs (For Your Information). They are written in a language much more informal than that used in the other, standard RFCs. Topics range from answers to common questions for new and experienced users to a suggested bibliography.

Finally, as the Internet has grown and technology has changed, some RFCs become unnecessary. These obsolete RFCs cannot be ignored, however. Frequently when a change is made to some RFC that causes a new one to obsolete others, the new RFC only contains explanations and motivations for the change. Understanding the model on which the whole facility is based may involve reading the original and subsequent RFCs on the topic.

RFCs and FYIs are available via FTP from many sources, including:

The nic.ddn.mil archive, as /rfc/rfc-xxxx.txt, where xxxx is the number of the RFC.

from ftp.uu.net, in the directory /RFC.

They're also available through mail by writing to service@nic.ddn.mil, with a Subject: line of send RFC-xxxx.TXT, again with xxxx being the RFC number.

"Knowledge is of two kinds. We know a subject ourselves, or we know where we can find information upon it." Samuel Johnson Letter to Lord Chesterfield February, 1755 a book of quotes said April 18, 1775 .. the book of Johnson's works said it's 1755; I'll go with the latter.

———— Conclusion

This guide is far from complete—the Internet changes on a daily (if not hourly) basis. However, this booklet should provide enough information to make the incredible breadth and complexity of the Internet a mite less imposing. Coupled with some exploration and experimentation, every user has the potential to be a competent net citizen, using the facilities that are available to their fullest.

You, the reader, are strongly encouraged to suggest improvements to any part of this booklet. If something was unclear, left you with doubts, or wasn't addressed, it should be fixed. If you find any problems, inaccuracies, spelling errors, etc., please report them to:

Brendan Kehoe
Department of Computer Science
Widener University
Chester, PA 19013

Internet: guide-bugs@cs.widener.edu
UUCP: ...!widener!guide-bugs

If you are interested in future updates to this guide (aside from normal new editions), discussion about information to be included or removed, etc., write to guide-request@cs.widener.edu to be placed on a mailing list for such things.

@dots is actually ` . . . ' "I've seed de first an de last @dots I seed de beginnin, en now I sees de endin." William Faulkner *The Sound & The Fury* April 8, 1928

Getting to Other Networks

Inter-connectivity has been and always will be one of the biggest goals in computer networking. The ultimate desire is to make it so one person can contact anyone else no matter where they are. A number of "gateways" between networks have been set up. They include:

AppleLink Quantum Services sells access to AppleLink, which is similar to QuantumLink for Commodore computers and PCLink for IBM PCs and compatibles. It also provides email access through the address user@applelink.apple.com.

ATTMail AT&T sells a commercial email service called ATTMail. Its users can be reached by writing to user@attmail.com.

BIX Users on BIX (the Byte Information eXchange) can be reached through the DAS gateway at user@cibix.das.net.

CompuServe (CI\$) To reach a user on the commercial service CompuServe, you must address the mail as xxxxx.xxx@compuserve.com, with xxxxx.xxx being their CompuServe user ID. Normally CompuServe ids are represented as being separated by a comma (like 71999,141); since

most mailers don't react well to having commas in addresses, it was changed to a period. For the above address, mail would be sent to 71999.141@compuserve.com.

EasyNet Digital sells a service called EasyNet; users that subscribe to it can be reached with the addresses `user@host.enet.dec.com` or `user%host.enet@decwrl.dec.com`.

FidoNet The FidoNet computer network can be reached by using a special addressing method. If John Smith is on the node 1:2/3.4 on FidoNet, his or her email address would be `john.smith@p4.f3.n2.z1.fidonet.org` (notice how the numbers fall in place?).

MCI Mail MCI also sells email accounts (similar to ATTMail). Users can be reached with `user@mcimail.com`.

PeaceNet Users on the PeaceNet network can be reached by writing to `user@igc.org`.

The Well Users on the service The Well can be reached by writing to `user@well.sf.ca.us`. The Well is directly connected to the Internet.

This table is far from complete. In addition to sites not being listed, some services are not (nor do they plan to be) accessible from the "outside" (like Prodigy); others, like GENie, are actively investigating the possibility of creating a gateway into their system. For the latest information, consult a list called the Inter-Network Mail Guide. It's available from a number of FTP sites, including UUNET; Anonymous FTP, for more information on getting a copy of it using anonymous FTP.

Retrieving Files via Email

For those who have a connection to the Internet, but cannot FTP, there do exist a few alternatives to get those files you so desperately need. When requesting files, it's imperative that you keep in mind the size of your request—odds are the other people who may be using your link won't be too receptive to sudden bursts of really heavy traffic on their normally sedate connection.

Archive Servers

An alternative to the currently well over-used FTPmail system is taking advantage of the many archive servers that are presently being maintained. These are programs that receive email messages that contain commands, and act on them. For example, sending an archive server the command help will usually yield, in the form of a piece of email, information on how to use the various commands that the server has available.

One such archive server is `service@nic.ddn.mil`. Maintained by the Network Information Center (NIC) in Chantilly, VA, the server is set up to make all of the information at the NIC available for people who don't have access to FTP. This also includes the WHOIS service (Whois). Some sample Subject: lines for queries to the NIC server are:

Subject: help Describes available commands.

Subject: rfc 822 Sends a copy of RFC-822.

Subject: rfc index Sends an index of the available RFCs.

Subject: netinfo domain-template.txt Sends a domain application.

Subject: whois widener Sends WHOIS information on `widener'.

More information on using their archive server can be obtained by writing to their server address `service@nic.ddn.mil` with a Subject: of help.

There are different "brands" of archive server, each with its own set of commands and services. Among them there often exists a common set of commands and services (e.g. index, help, etc). Be that as it may, one should always consult the individual help for a specific server before assuming the syntax—100K surprises can be hard on a system.

FTP-by-Mail Servers Some systems offer people the ability to receive files through a mock-FTP interface via email. Anonymous FTP for a general overview of how to FTP. The effects of providing such a service varies, although a rule of thumb is that it will probably use a substantial amount of the available resources on a system.

The "original" FTP-by-Mail service, BITFTP, is available to BITNET users from the Princeton node PUCC. It was once accessible to anyone, but had to be closed out to non-BITNET users because of the heavy load on the system.

In response to this closure, Paul Vixie designed and installed a system called FTPmail on one of Digital's gateway computers, decwrl.dec.com. Write to ftpmail@decwrl.dec.com with help in the body of the letter for instructions on its use. The software is undergoing constant development; once it reaches a stable state, other sites will be encouraged to adopt it and provide the service also.

Newsgroup Creation

Everyone has the opportunity to make a Call For Votes on the Usenet and attempt to create a newsgroup that he/she feels would be of benefit to the general readership. The rules governing newsgroup creation have evolved over the years into a generally accepted method. They only govern the "world" groups; they aren't applicable to regional or other alternative hierarchies.

Discussion

A discussion must first take place to address issues like the naming of the group, where in the group tree it should go (e.g. rec.sports.koosh vs rec.games.koosh?), and whether or not it should be created in the first place. The formal Request For Discussion (RFD) should be posted to news.announce.newgroups, along with any other groups or mailing lists at all related to the proposed topic. news.announce.newgroups is moderated. You should place it first in the Newsgroups: header, so that it will get mailed to the moderator only. The article won't be immediately posted to the other newsgroups listed; rather, it will give you the opportunity to have the moderator correct any inconsistencies or mistakes in your RFD. He or she will take care of posting it to the newsgroups you indicated. Also the Followup-To: header will be set so that the actual discussion takes place only in news.groups. If a user has difficulty posting to a moderated group, he or she may mail submissions intended for news.announce.newgroups to the address announce-newgroups@rpi.edu.

The final name and charter of the group, and whether it will be moderated or unmoderated, will be determined during the discussion period. If it's to be moderated, the discussion will also decide who the moderator will be. If there's no general agreement on these points among those in favor of a new group at the end of 30 days, the discussion will be taken into mail rather than continued posting to news.groups; that way, the proponents of the group can iron out their differences and come back with a proper proposal, and make a new Request For Discussion.

Voting After the discussion period (which is mandatory), if it's been determined that a new group really is desired, a name and charter are agreed upon, and it's been determined whether the group will be moderated (and by whom), a Call For Votes (CFV) should be posted to

news.announce.newgroups, along with any other groups that the original Request For Discussion was posted to. The CFV should be posted (or mailed to the news.announce.newgroups moderator) as soon as possible after the discussion ends (to keep it fresh in everyone's mind).

The Call for Votes should include clear instructions on how to cast a vote. It's important that it be clearly explained how to both vote for and against a group (and be of equivalent difficulty or ease). If it's easier for you or your administrator, two separate addresses can be used to mail yes and no votes to, providing that they're on the same machine. Regardless of the method, everyone must have a very specific idea of how to get his/her vote counted.

The voting period can last between 21 and 31 days, no matter what the preliminary results of the vote are. A vote can't be called off simply because 400 "no" votes have come in and only two "yes" votes. The Call for Votes should include the exact date that the voting period will end—only those votes arriving on the vote-taker's machine before this date can be counted.

To keep awareness high, the CFV can be repeated during the vote, provided that it gives the same clear, unbiased instructions for casting a vote as the original; it also has to be the same proposal as was first posted. The charter can't change in mid-vote. Also, votes that're posted don't count—only those that were mailed to the vote-taker can be tallied.

Partial results should never be included; only a statement of the specific proposal, that a vote is in progress on it, and how to cast a vote. A mass acknowledgement ("Mass ACK" or "Vote ACK") is permitted; however, it must be presented in a way that gives no indication of which way a person voted. One way to avoid this is to create one large list of everyone who's voted, and sort it in alphabetical order. It should not be two sorted lists (of the yes and no votes, respectively).

Every vote is autonomous. The votes for or against one group can't be transferred to another, similar proposal. A vote can only count for the exact proposal that it was a response to. In particular, a vote for or against a newsgroup under one name can't be counted as a vote for or against another group with a different name or charter, a different moderated/unmoderated status, or, if it's moderated, a different moderator or set of moderators. Whew!

Finally, the vote has to be explicit; they should be of the form I vote for the group foo.bar as proposed or I vote against the group foo.bar as proposed. The wording doesn't have to be exact, your intention just has to be clear.

The Result of a Vote

At the end of the voting period, the vote-taker has to post (to news.announce.newgroups) the tally and email addresses of the votes received. Again, it can also be posted to any of the groups listed in the original CFV. The tally should make clear which way a person voted, so the results can be verified if it proves necessary to do so.

After the vote result is posted to news.announce.newgroups, there is a mandatory five-day waiting period. This affords everyone the opportunity to correct any errors or inconsistencies in the voter list or the voting procedure.

Creation of the Group

If, after the waiting period, there are no serious objections that might invalidate the vote, the vote is put to the "water test." If there were 100 more valid YES/create votes than NO/don't create votes, and at least two-thirds of the total number of votes are in favor of creation, then a newgroup

control message can be sent out (often by the moderator of news.announce.newgroups). If the 100-vote margin or the two-thirds percentage isn't met, the group has failed and can't be created.

If the proposal failed, all is not lost—after a six-month waiting period (a "cooling down"), a new Request For Discussion can be posted to news.groups, and the whole process can start over again. If after a couple of tries it becomes obvious that the group is not wanted or needed, the vote-taker should humbly step back and accept the opinion of the majority. (As life goes, so goes Usenet.)

Glossary

This glossary is only a tiny subset of all of the various terms and other things that people regularly use on The Net. For a more complete (and very entertaining) reference, it's suggested you get a copy of The New Hacker's Dictionary, which is based on a VERY large text file called the Jargon File. Edited by Eric Raymond (eric@snark.thyrsus.com), it is available from the MIT Press, Cambridge, Massachusetts, 02142; its ISBN number is 0-262-68069-6. Also see RFC-1208, A Glossary of Networking Terms.

:-) This odd symbol is one of the ways a person can portray "mood" in the very flat medium of computers—by using "smilies." This is 'metacommunication', and there are literally hundreds of them, from the obvious to the obscure. This particular example expresses "happiness." Don't see it? Tilt your head to the left 90 degrees. Smilies are also used to denote sarcasm.

Network addresses are usually of two types:

the physical or hardware address of a network interface card; for ethernet this 48-bit address might be 0260.8C00.7666. The hardware address is used to forward packets within a physical network. Fortunately, network users do not have to be concerned about hardware addresses since they are automatically handled by the networking software.

The logical or Internet address is used to facilitate moving data between physical networks. The 32-bit Internet address is made up of a network number, a subnetwork number, and a host number. Each host computer on the Internet, has a unique address. For example, all Internet addresses at Colorado State have a network number of 129.82, a subnet number in the range of 1-254, and a host number in the range of 1-254. All Internet hosts have a numeric address and an English-style name. For example, the Internet address for UCC's CYBER 840 is 129.82.103.96; its Internet name is csugreen.UCC.ColoState.EDU.

address resolution

Conversion of an Internet address to the corresponding physical address.

On an ethernet, resolution requires broadcasting on the local area network.

administrivia Administrative tasks, most often related to the maintenance of mailing lists, digests, news gateways, etc.

anonymous FTP Also known as "anon FTP"; a service provided to make files available to the general Internet community—Anonymous FTP.

ANSI The American National Standards Institute disseminates basic standards like ASCII, and acts as the United States' delegate to the ISO. Standards can be ordered from ANSI by writing to the ANSI Sales Department, 1430 Broadway, New York, NY 10018, or by telephoning (212) 354-3300.

archie A service which provides lookups for packages in a database of the offerings of countless of anonymous FTP sites. archie for a full description.

archive server An email-based file transfer facility offered by some systems.

ARPA (Advanced Research Projects Agency) Former name of DARPA, the government agency that funded ARPAnet and later the DARPA Internet.

ARPAnet A pioneering long haul network funded by ARPA. It served as the basis for early networking research as well as a central backbone during the development of the Internet. The ARPAnet consisted of individual packet switching computers interconnected by leased lines. The ARPAnet no longer exists as a singular entity.

asynchronous Transmission by individual bytes, not related to specific timing on the transmitting end.

auto-magic Something which happens pseudo-automatically, and is usually too complex to go into any further than to say it happens "auto-magically."

backbone A high-speed connection within a network that connects shorter, usually slower circuits. Also used in reference to a system that acts as a "hub" for activity (although those are becoming much less prevalent now than they were ten years ago).

bandwidth The capacity of a medium to transmit a signal. More informally, the mythical "size" of The Net, and its ability to carry the files and messages of those that use it. Some view certain kinds of traffic (FTPing hundreds of graphics images, for example) as a "waste of bandwidth" and look down upon them.

BITNET (Because It's Time Network)

An NJE-based international educational network.

bounce

The return of a piece of mail because of an error in its delivery.

btw

An abbreviation for "by the way."

CFV (Call For Votes) Initiates the voting period for a Usenet newsgroup. At least one (occasionally two or more) email address is customarily included as a repository for the votes. See Newsgroup Creation for a full description of the Usenet voting process.

ClariNews

The fee-based Usenet newsfeed available from ClariNet Communications.

client The user of a network service; also used to describe a computer that relies upon another for some or all of its resources.

Cyberspace

A term coined by William Gibson in his fantasy novel Neuromancer to describe the "world" of computers, and the society that gathers around them.

datagram The basic unit of information passed across the Internet. It contains a source and destination address along with data. Large messages are broken down into a sequence of IP datagrams.

disassembling

Converting a binary program into human-readable machine language code.

DNS (Domain Name System)

The method used to convert Internet names to their corresponding Internet numbers.

domain A part of the naming hierarchy. Syntactically, a domain name consists of a sequence of names or other words separated by dots.

dotted quad A set of four numbers connected with periods that make up an Internet address; for example, 147.31.254.130.

email The vernacular abbreviation for electronic mail.

email address The UUCP or domain-based address that a user is referred to with. For example, the author's address is `brendan@cs.widener.edu`.

ethernet A 10-million bit per second networking scheme originally developed by Xerox Corporation. Ethernet is widely used for LANs because it can network a wide variety of computers, it is not proprietary, and components are widely available from many commercial sources.

FDDI (Fiber Distributed Data Interface) An emerging standard for network technology based on fiber optics that has been established by ANSI. FDDI specifies a 100-million bit per second data rate. The access control mechanism uses token ring technology.

flame A piece of mail or a Usenet posting which is violently argumentative.

FQDN (Fully Qualified Domain Name) The FQDN is the full site name of a system, rather than just its hostname. For example, the system lisa at Widener University has a FQDN of `lisa.cs.widener.edu`.

FTP (File Transfer Protocol) The Internet standard high-level protocol for transferring files from one computer to another.

FYI An abbreviation for the phrase "for your information." There is also a series of RFCs put out by the Network Information Center called FYIs; they address common questions of new users and many other useful things. RFCs for instructions on retrieving FYIs.

gateway A special-purpose dedicated computer that attaches to two or more networks and routes packets from one network to the other. In particular, an Internet gateway routes IP datagrams among the networks it connects. Gateways route packets to other gateways until they can be delivered to the final destination directly across one physical network.

header The portion of a packet, preceding the actual data, containing source and destination addresses and error-checking fields. Also part of a message or news article.

hostname The name given to a machine. (See also FQDN.)

IMHO (In My Humble Opinion) This usually accompanies a statement that may bring about personal offense or strong disagreement.

Internet A concatenation of many individual TCP/IP campus, state, regional, and national networks (such as NSFnet, ARPAnet, and Milnet) into one single logical network all sharing a common addressing scheme.

Internet number The dotted-quad address used to specify a certain system. The Internet number for the site cs.widener.edu is 147.31.254.130. A resolver is used to translate between hostnames and Internet addresses.

interoperate The ability of multi-vendor computers to work together using a common set of protocols. With interoperability, PCs, Macs, Suns,

Dec VAXen, CDC Cybers, etc, all work together allowing one host computer to communicate with and take advantage of the resources of another.

ISO (International Organization for Standardization)
Coordinator of the main networking standards that are put into use today.

kernel The level of an operating system or networking system that contains the system-level commands or all of the functions hidden from the user. In a Unix system, the kernel is a program that contains the device drivers, the memory management routines, the scheduler, and system calls. This program is always running while the system is operating.

LAN (Local Area Network) Any physical network technology that operates at high speed over short distances (up to a few thousand meters).

mail gateway A machine that connects to two or more electronic mail systems (especially dissimilar mail systems on two different networks) and transfers mail messages among them.

mailing list A possibly moderated discussion group, distributed via email from a central computer maintaining the list of people involved in the discussion.

mail path A series of machine names used to direct electronic mail from one user to another.

medium The material used to support the transmission of data. This can be copper wire, coaxial cable, optical fiber, or electromagnetic wave (as in microwave).

multiplex The division of a single transmission medium into multiple logical channels supporting many simultaneous sessions. For example, one

network may have simultaneous FTP, telnet, rlogin, and SMTP connections, all going at the same time.

net.citizen An inhabitant of Cyberspace. One usually tries to be a good net.citizen, lest one be flamed.

netiquette A pun on "etiquette"; proper behavior on The Net. Usenet Netiquette.

network A group of machines connected together so they can transmit information to one another. There are two kinds of networks: local networks and remote networks.

NFS (Network File System) A method developed by Sun Microsystems to allow computers to share files across a network in a way that makes them appear as if they're "local" to the system.

NIC
The Network Information Center.

node
A computer that is attached to a network; also called a host.

NSFnet The national backbone network, funded by the National Science Foundation and operated by the Merit Corporation, used to interconnect regional (mid-level) networks such as WestNet to one another.

packet The unit of data sent across a packet switching network. The term is used loosely. While some Internet literature uses it to refer specifically to data sent across a physical network, other literature views the Internet as a packet switching network and describes IP datagrams as packets.

polling Connecting to another system to check for things like mail or news.

postmaster The person responsible for taking care of mail problems, answering queries about users, and other related work at a site.

protocols A formal description of message formats and the rules two computers must follow to exchange those messages. Protocols can describe low-level details of machine-to-machine interfaces (e.g., the order in which bits and bytes are sent across a wire) or high-level exchanges between allocation programs (e.g., the way in which two programs transfer a file across the Internet).

recursion The facility of a programming language to be able to call functions from within themselves.

resolve Translate an Internet name into its equivalent IP address or other DNS information.

RFD (Request For Discussion) Usually a two- to three-week period in which the particulars of newsgroup creation are battled out.

route The path that network traffic takes from its source to its destination.

router A dedicated computer (or other device) that sends packets from one place to another, paying attention to the current state of the network.

RTFM (Read The Fantastic Manual). This anacronym is often used when someone asks a simple or common question. The word 'Fantastic' is usually replaced with one much more vulgar.

SMTP (Simple Mail Transfer Protocol) The Internet standard protocol for transferring electronic mail messages from one computer to another. SMTP specifies how two mail systems interact and the format of control messages they exchange to transfer mail.

server

A computer that shares its resources, such as printers and files, with other computers on the network. An example of this is a Network File System (NFS) server which shares its disk space with other computers.

signal-to-noise ratio When used in reference to Usenet activity, signal-to-noise ratio describes the relation between amount of actual information in a discussion, compared to their quantity. More often than not, there's substantial activity in a newsgroup, but a very small number of those articles actually contain anything useful.

signature The small, usually four-line message at the bottom of a piece of email or a Usenet article. In Unix, it's added by creating a file `..signature` in the user's home directory. Large signatures are a no-no.

summarize To encapsulate a number of responses into one coherent, usable message. Often done on controlled mailing lists or active newsgroups, to help reduce bandwidth.

synchronous Data communications in which transmissions are sent at a fixed rate, with the sending and receiving devices synchronized.

TCP/IP (Transmission Control Protocol/Internet Protocol)

A set of protocols, resulting from ARPA efforts, used by the Internet to support services such as remote login (telnet), file transfer (FTP) and mail (SMTP).

telnet The Internet standard protocol for remote terminal connection service. Telnet allows a user at one site to interact with a remote timesharing system at another site as if the user's terminal were connected directly to the remote computer.

terminal server A small, specialized, networked computer that connects many terminals to a LAN through one network connection. Any user on the network can then connect to various network hosts.

TeX

A free typesetting system by Donald Knuth.

twisted pair Cable made up of a pair of insulated copper wires wrapped around each other to cancel the effects of electrical noise.

UUCP (Unix to Unix Copy Program) A store-and-forward system, primarily for Unix systems but currently supported on other platforms (e.g. VMS and personal computers).

WAN (Wide-Area Network)

A network spanning hundreds or thousands of miles.

workstation

A networked personal computing device with more power than a standard IBM PC or Macintosh. Typically, a workstation has an operating system such as unix that is capable of running several tasks at the same time. It has several megabytes of memory and a large, high-resolution display. Examples are Sun workstations and Digital DECstations.

worm A computer program which replicates itself. The Internet worm (The Internet Worm) was perhaps the most famous; it successfully (and accidentally) duplicated itself on systems across the Internet.

wrt

With respect to.

"I hate definitions."

Benjamin Disraeli

Vivian Grey, bk i chap ii

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"And all else is literature."

Paul Verlaine

The Sun, New York

While he was city editor in 1873—1890.

— Bill Walther, Carleton University, Ottawa, Canada

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ART OF THE INTERNET ***

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